Gastrointestinal system 14

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General principles

A key requirement of being a good doctor is the ability to make a robust diagnosis. Making a diagnosis requires identifying departure from normal and involves answering two key questions:

1. Where is the problem (i.e. the anatomical site)?
2. What is the problem (i.e. nature of disease)?

A good history and a focused examination are the keys to making a diagnosis/differential diagnoses, and this is particularly true when patients present with diseases involving the gastrointestinal (GI) system. This chapter covers the key GI symptoms, how to take a focused history and the diseases likely to be associated with such symptoms. It then proceeds to cover examination technique for assessing the GI system, followed by a brief review of investigations required to confirm a diagnosis.

Anatomy of the GI tract

The human gastrointestinal tract can be seen as a system of serially connected tubular organs approximately 8 metres in length (Fig. 14.1) including mouth, oesophagus, stomach, duodenum, jejunum, ileum, colon, rectum and anus which, together with its connected secretory glands including the salivary glands, pancreas and biliary tree within the liver, controls the passage, processing, absorption and elimination of food. Symptoms of gastrointestinal (GI) disorders usually arise owing to an alteration in this transit, or an alteration in the function of this system, which includes absorption, processing and elimination of food.

The site of disease may lie in the lumen, its wall (including the mucosa, submucosa, muscle layer or serosa); involve the secretions that aid digestion (biliary and pancreatic diseases); and even lie outside the GI tract (extra luminal), affecting the GI system either through extrinsic compression or through neuroendocrine effects as seen in patients with diabetes or through electrolyte disturbances as in renal diseases. The liver, biliary system and pancreas, which are part of the GI system, are involved in aiding digestion. In addition, the liver is an important organ playing a key role in immunity as well as in metabolism of food and drugs, and also elimination of waste products such as bilirubin (a byproduct of haemoglobin breakdown) into bile.

Finally, from the standpoint of systematic history taking and examination, the kidneys, groin and genitalia are also considered in this chapter.

There are a large number of organs within GI tract, so to make it easier to diagnose the site of a problem, it may be helpful to consider the GI system composed broadly of three subsystems as follows:

1. Upper GI tract: This includes the oral cavity, oesophagus, stomach and proximal duodenum, traditionally as far as the ligament of Trietz: symptoms of diseases involving this section include dysphagia, odynophagia, reflux, heartburn, epigastric pain, and vomiting.
2. Lower GI tract: This includes small intestines, including distal duodenum, jejunum, ileum, colon, rectum and anus: common symptoms from diseases involving this section include change in bowel habit (this can range from constipation to diarrhoea, or a combination), bleeding per rectum, abdominal pain and distension.
3. Hepatopancreatico-biliary (HPB): This section includes liver, biliary tree including gall bladder and pancreas: diseases involving this section will present with symptoms that include jaundice and abdominal pain with biliary pain classically described as right upper quadrant abdominal pain radiating to the right shoulder. However, not all HPB diseases may be associated with jaundice and pure liver diseases rarely cause pain.

In addition to the above symptoms, GI diseases from all the three subsystems as above may present with diffuse or poorly localized abdominal pain, weight loss, loss of appetite and nausea. These symptoms, although of little localising value, may be useful in assessing the severity of underlying disease.

Diseases of the GI tract

A wide variety of diseases are associated with different sites of the GI tract. For simplicity, one way of classifying these is based on the underlying causes as shown in Box 14.1. However, in clinical medicine

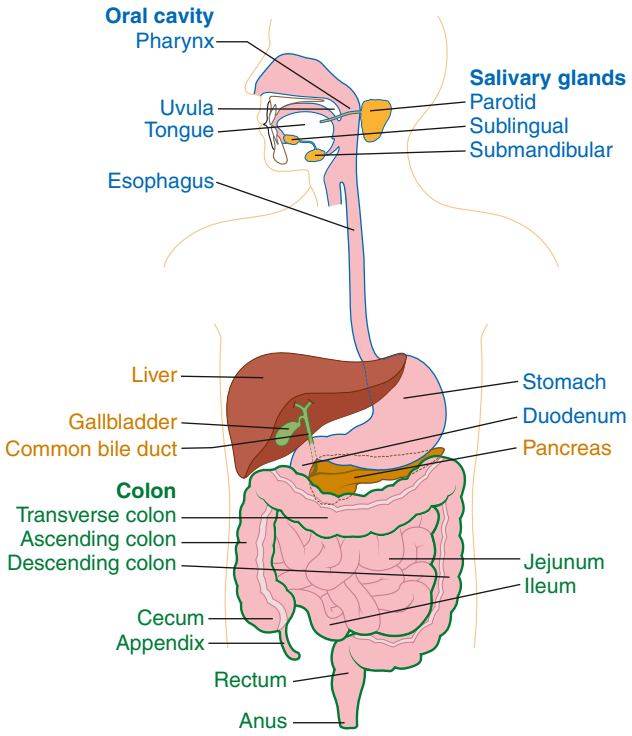


Figure 14.1 Anatomy of the GI tract includes three sub-systems: upper gastrointestinal (UGI), includes all organs labelled in blue colour; hepatopancreaticobiliary (HPB), includes all organs labelled in orange colour; lower gastrointestinal (LGI), includes all organs labelled in green colour.

Box 14.1 The types of diseases seen in gastrointestinal system

- V-Vascular: Ischaemic bowel, hepatic infarction
- I-Inflammation/infection: Inflammatory bowel disease, viral hepatitis (A, B, C, D, E)
- N-Neoplasms: Cancers of oesophagus, stomach, small bowel, colon, liver, pancreas, biliary tree
- D-Drug (toxins)/degenerative/dysmotility: Drug-induced diarrhoea/hepatitis, irritable bowel syndrome
- I-Iatrogenic/idiopathic: Perforation of GI tract post endoscopy, bleed post liver biopsy/trauma
- C-Congenital/inherited/developmental: Pyloric stenosis, Hirschsprung's disease
- A-Autoimmune/allergy: Autoimmune hepatitis, coeliac disease
- T-Trauma
- E-Endocrine/metabolic/environmental: Diarrhoea caused by diabetic neuropathy

it is important to be aware of the common diseases and to include these in the differential diagnosis before considering rare ones. Table 14.1 includes list of common GI diseases.

Table 14.1 Common GI diseases

Site	Diseases
Oesophagus	Oesophagitis, gastroesophageal reflux disease, hiatus hernia, oesophageal ulcer, benign oesophageal stricture, Barratt's oesophagus, oesophageal cancer, oesophageal varices
Stomach	Gastritis, gastric ulcer, gastric cancer, gastric varices
Small bowel	Duodenitis, duodenal ulcer, coeliac disease, Crohn's disease, lymphoma
Colorectal	Diverticulosis, ulcerative colitis, other colitis including infective, colonic/rectal polyps and cancers
Hepatic	Acute hepatitis, liver cirrhosis, liver abscess, liver cancers (primary/secondary)
Pancreatico-biliary	Gall bladder and CBD stones, cholangitis, cancer gall bladder and biliary tree, acute and chronic pancreatitis, cancer of the pancreas

Symptoms of gastrointestinal disease

The common symptoms of GI and abdominal disease are listed in Box 14.2 and are discussed individually below.

An important aspect to remember while taking a history of GI symptoms is that certain symptoms and signs are associated with a higher likelihood of an underlying cancer, especially if recent in onset (i.e. less than 3 months in duration) and in an older patient (>55 years of age). These require urgent assessment, including referral to an appropriate specialist; to highlight this urgency such symptoms are referred to as red flags. The relevant red flags for GI diseases are shown in Box 14.3.

Box 14.2 Common symptoms of gastrointestinal and abdominal disease

- Dysphagia and odynophagia
- Heartburn and reflux
- Indigestion
- Flatulence
- Vomiting
- Anorexia
- Constipation
- Diarrhoea
- Alteration of bowel pattern
- Abdominal pain
- Abdominal distension
- Weight loss
- Haematemesis
- Rectal bleeding
- Melaena
- Jaundice
- Itching
- Urinary symptoms

Box 14.3 Gastrointestinal red flag symptoms

- Dysphagia
- Dyspepsia: new onset lasting for weeks
- Abdominal pain: new onset lasting for weeks
- Change in bowel habit: diarrhoea or constipation, sense of incomplete evacuation
- Rectal bleeding
- Weight loss
- Loss of appetite
- New onset iron deficiency anaemia in non-menstruating women or in men

Dysphagia (and odynophagia)

Dysphagia is defined as difficulty in swallowing. Patients may report an array of symptoms ranging from an awareness of something sticking in the throat or chest to an inability to swallow.

Dysphagia is a red flag symptom because it can be associated with oesophageal cancer or cancer of the gastric cardia or fundus. Other symptoms may accompany dysphagia and it is important to be familiar with the physiology of swallowing to understand the localizing value of these accompanying symptoms to help make an accurate diagnosis.

Swallowing is a complex movement composed of two phases: the initial oropharyngeal phase and the oesophageal phase (Fig. 14.2). The oropharyngeal phase involves multiple actions that happen synchronously, which are designed to allow safe passage of food bolus from the mouth into the oesophagus, without it going into the trachea (aspiration) or refluxing in the nasal cavity. As the food is moved to the back of the mouth by the tongue, the tongue closes the oral cavity, the soft palate closes the nasal cavity and the epiglottis closes the larynx, so the food is safely propelled into the proximal oesophagus. Diseases affecting the oropharyngeal phase of swallowing include local diseases of the throat, such as cancer of the throat, which may present with symptoms of dysphagia. More commonly neurological conditions, such as stroke, affect swallowing and patients may present with symptoms of choking, recurrent aspirations with frequent chest infections and, rarely, reflux of food through the nose.

The second phase of swallowing is the oesophageal phase, initiated on entry of food bolus into the proximal oesophagus; it involves movement of food down the oesophagus by peristalsis aided by gravity

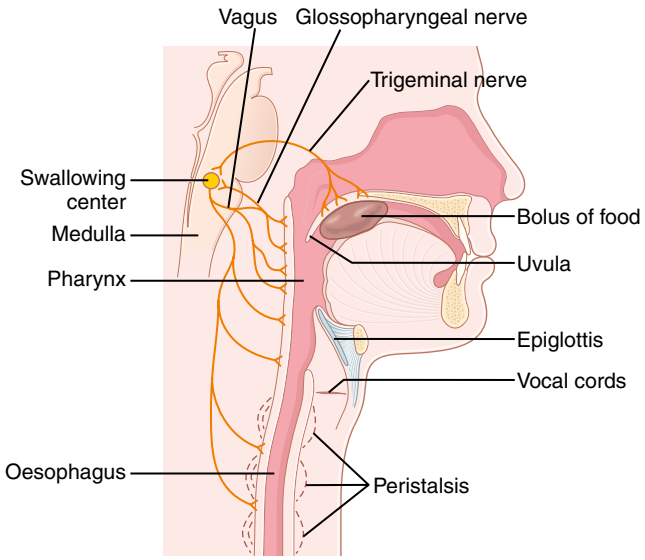


Figure 14.2 Mechanism of swallowing. (From Hall JE, Hall ME, eds. Guyton and Hall Textbook of Medical Physiology, ed 14. Figure 64.1. 2021. Elsevier Inc.)

Box 14.4 Causes of dysphagia

Obstructive

- Intraluminal: Foreign body
- Within walls: Stricture, cancer
- Extraluminal: Goitre, thymoma, mediastinal lymphadenopathy, compression from enlarged aorta or heart

Non-obstructive

- Oesophageal: Achalasia cardia, eosinophilic oesophagitis, oesophageal dysmotility
- Systemic: Stroke, multiple sclerosis, myasthenia gravis

Box 14.5 Key features in history of dysphagia

- Onset: Sudden versus gradual
- Duration of symptoms: years or months or days
- Is it worse with solids than liquids?
- Difficulty in swallowing saliva
- Accompanied with reflux
- Loss of weight, loss of appetite, iron deficiency anaemia

and relaxation of the lower oesophageal sphincter (LOS) to allow passage of food into the stomach. Intact tone of the LOS is required to prevent reflux of gastric content into the oesophagus. Problems in the oesophageal phase can result in symptoms of dysphagia, reflux, heartburn and chest pain.

Box 14.4 shows the causes of dysphagia which can be broadly classified into obstructive and non-obstructive. Obstructive dysphagia may be caused by problems, firstly within the lumen such as foreign body, secondly within the wall, such as benign or malignant oesophageal strictures or thirdly extra-luminal compression of the lumen such as from goitre, thymoma, enlarged mediastinal lymph nodes or aortic aneurysm. Non-obstructive causes broadly refer to conditions that may affect oesophageal peristalsis, which can be caused by disorders within the oesophageal muscles—referred to as oesophageal dysmotility, the most prominent of these being localized to oesophagus such as achalasia of the cardia or eosinophilic oesophagitis or connective diseases such as systemic sclerosis. Dysmotility may also be owing to neurological disorders affecting the neuronal control of peristalsis, which include systemic diseases such as stroke, multiple sclerosis or myasthenia gravis.

Classifying dysphagia into obstructive and non-obstructive/dysmotility is relevant because these two types can be distinguished on the basis of an astute history, which should include questions in Box 14.5.

The presence of dysphagia worse to solids would suggest an obstructive dysphagia; if it is progressively worse over a short time and accompanied with weight loss, then oesophageal malignancy should be suspected. A benign stricture (or rarely an oesophageal pouch) may follow the same pattern, but much less rapidly and only delayed weight loss. Neurogenic dysphagia may present with greater difficulty in swallowing with liquids than solids and, more importantly, may often be associated with aspiration or coughing and seen in patients with stroke, motor neuron disease and myotonic dystrophy.

In contrast, odynophagia is painful swallowing owing to inflammation in the oropharynx or proximal oesophagus, which may occur with or without dysphagia. Odynophagia indicates inflammation/infection of the oralpharyngeal or oesophageal mucosa, classically candida oesophagitis, which is commonly associated with recent therapy with steroids or antibiotics and the presence of diabetes. However, diffuse oesophageal candidiasis may be associated with immuno-compromised states, such as after chemotherapy or HIV infection.

Heartburn

Heartburn is a burning sensation perceived in the chest or neck. It is caused by acid reflux from the stomach into the oesophagus and seen in patients with gastro-oesophageal reflux disease (GORD). Occasionally it is difficult to distinguish from angina pectoris. It is reported as present particularly at night when the patient lies flat in bed or after bending or stooping when abdominal pressure is increased. Heartburn may be exacerbated by dietary intake (such as alcohol or very spicy or fatty foods, which lower tone of the lower oesophageal sphincter) and certain medications (such as bisphosphonates). However, not all patients with reflux will have symptoms of heartburn.

Reflux

Reflux or regurgitation is a symptom which refers to non-acidic fluid or bile regurgitation into the mouth, causing a bitter taste and a disagreeable sensation retrosternally. It is suggestive of underlying GORD, but not all patients with GORD have reflux symptoms.

Indigestion (dyspepsia)

Dyspepsia, a medical term for indigestion, includes a wide variety of symptoms, such as epigastric pain, heartburn, distension, nausea or 'an acid feeling' occurring after eating or drinking, bloating and belching. It may be associated with diseases of the upper GI tract such as *Helicobacter* infection of stomach or duodenum, peptic ulcer disease and GORD. Upper GI malignancy should be excluded in older patients who present with a new onset of dyspepsia. It can also be associated with diseases of the biliary tree, such as chronic cholecystitis (inflammation of the gall bladder), or just the presence of gallstones.

Vomiting

Vomiting is forceful expulsion of gastric contents through the mouth or nose. It occurs either

consequent upon an obstruction in the transit of food through the GI tract or as a neurogenic response, in turn triggered either by chemoreceptors in the brainstem or as a reflex from irritation of the stomach. Vomiting consists of a phase of nausea, followed by hypersalivation, pallor, sweating and hyperventilation. Retching, an involuntary effort to vomit, then occurs, followed by expulsion of gastric contents through the mouth and sometimes through the nose. Most nausea and vomiting of GI origin are associated with local discomfort in the abdomen. Non-GI disease, such as raised intracranial pressure or metabolic disturbance, should be suspected if vomiting is not associated with food or any preceding abdominal discomfort.

It is important to take a history of the content of the vomit, its relationship with meals, frequency and association with other symptoms such as abdominal pain or constipation. The content of the vomit varies with the nature of the food ingested and it is important to note the absence or presence of bile or blood. In pyloric stenosis, the vomit is usually copious and sour-smelling. It frequently contains recognizable food eaten many hours before and exhibits froth on the surface after standing, but no bile. Faeculent vomit, characteristic of advanced intestinal obstruction, is brown in colour, and has a faecal odour. Vomit containing formed faeces is rare and may indicate a fistula between stomach and transverse colon, usually from an invading colonic carcinoma.

Haematemesis

Haematemesis, which is vomiting of blood, results from bleeding from lesions in the upper GI tract. Brisk haematemesis, such as from a large vessel in a gastric or duodenal ulcer, will cause the patient to vomit large volumes of pure blood. In addition this will often be accompanied with passage of black tarry stools, known as melaena (see below). Vomiting of dark-stained liquid is referred as coffee ground vomiting. It is owing to blood that lies in the acidic gastric juice for a period of time, during which it turns black and may be vomited looking like ground coffee. It is important to distinguish this from haemoptysis, which is coughing up blood and may be mixed with sputum.

Melaena

Melaena, the passage of black, tarry stool, is often described as the 5s': schwartz (black), sticky, smelly shiny, stool. The black colour is owing to blood as it gets denatured (digested), as it passes through the acidic milieu in stomach and long transit through the small bowel. It usually indicates bleeding in the upper GI tract. Less commonly, a small bowel bleed proximal to the ileo-caecal valve may produce melaena and, very occasionally, the bleeding source may be in the right colon, such as a large caecal polyp or cancer, with altered colour and partial digestion facilitated by slow transit.

Box 14.6 Characteristics of pain from frequent and important causes (see also Fig. 14.7)

- Peptic ulcer: Epigastric, burning or gnawing, radiates through to back, meal related, wakes the patient, relieved by antacid
- Gastric cancer: Epigastric, severe, partly meal related, not relieved by antacid
- Pancreatic: High epigastric, severe, felt front-to-back, immediately after eating, relieved by sitting forward
- Midgut: Periumbilical, colicky, some relation to meals
- Lower gut: Periumbilical or suprapubic, colicky, some relief from bowel action
- Biliary: Right upper quadrant, severe, colicky (but over a long period of time), radiates to right shoulder, accompanied by nausea
- Renal colic: Loin-to-groin, colicky, very severe, accompanied by nausea
- Functional: Anywhere in the abdomen, colicky, accompanied by bloating, relieved by bowel action

Abdominal pain

Abdominal pain is one of the most common GI symptoms and can be produced by a wide variety of underlying diseases. As with any pain, it is important to characterize it in terms of its **s**ite, **o**nset, **c**haracter, areas of **r**adiation, **t**iming including duration and frequency, **e**xacerbating and relieving factors and associated features (the acronym SOCRATES is commonly used in practice as an aide-mémoire).

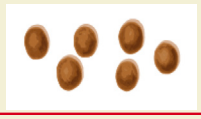
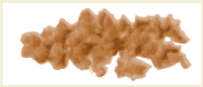
Acute abdominal pain is also discussed in Chapter 9 because it may be associated with serious life-threatening conditions and requires a distinct approach. The particular characteristics of pain from certain frequent and important causes are given in Box 14.6. Abdominal pain often radiates or spreads from one site to another. The pattern of radiation can indicate specific diseases. Pancreatic pain classically radiates through to the back and is relieved on bending forward, whereas pain from passage of ureteric stone radiates from loin to groin on the respective side.

It is important to be aware of referred pain in the abdomen when the pain is perceived away from the organ involved. The diaphragm is innervated by the phrenic nerve which derives from the C3, 4,5 nerve roots. Hence, pain may arise from diaphragmatic irritation owing to an inflamed gall bladder in cholecystitis or a liver abscess abutting the right diaphragm may be referred to the right shoulder, whereas pain from splenic pathology may be referred to the left shoulder. Pain from renal diseases is often referred to the lumbar area anteriorly or posteriorly.

Abdominal pain may also be owing to causes that are not specifically in the abdomen such as metabolic disorders including diabetic ketoacidosis, porphyria or lead poisoning.

Box 14.7 Bristol stool chart

Bristol Stool Chart

Type 1		Separate hard lumps, like nuts (hard to pass)
Type 2		Sausage-shaped but lumpy
Type 3		Like a sausage but with cracks on the surface
Type 4		Like a sausage or snake, smooth and soft
Type 5		Soft blobs with clear-cut edges
Type 6		Fluffy pieces with ragged edges, a mushy stool
Type 7		Watery, no solid pieces. **Entirely Liquid**



First published: Lewis SJ, Heaton KW. Stool form scale as a useful guide to intestinal transit time. *Scand J Gastroenterol.* 1997;32(9):920-924.

Source: https://www.nice.org.uk/guidance/cg99/resources/cg99-constipation-in-children-and-young-people-bristol-stool-chart-2

Abdominal distension

Abdominal distension is classically caused by one of the 'Five Fs': fluid (mainly ascites), fat (obesity), foetus (pregnancy), flatus and faeces. Consider whether the distension is symmetrical or asymmetrical. Common causes of symmetrical abdominal distension are obesity and ascites. Common causes of ascites include decompensated liver cirrhosis, congestive heart failure, nephrotic syndrome and abdominal malignancies.

Common causes of asymmetrical abdominal distension include marked enlargement of the major abdominal organs (hepatomegaly, splenomegaly) or the presence of a large mass lesion as seen in intra-abdominal cancers, lymphomas or peritoneal malignancies.

Change in bowel habit

Change in bowel habit is a broad term that includes diarrhoea, constipation, as well as a combination, where patients may report alternating diarrhoea and constipation. The frequency of 'normal' bowel action varies greatly from person to person and within different parts of the world. In high-income countries, with a greater ingestion of processed food, the statistical norm varies between three bowel actions per day to three per week. It is important to use a Bristol Stool chart (Box 14.7) to help objectively document the character of the stool, especially while monitoring response to an intervention or therapy.

Constipation

Constipation refers to a reduction in the volume or frequency of stool and/or an increase in the hardness of stool. In clinical practice, the passage of formed stool less frequently than three times per week is usually taken to indicate an abnormality of bowel frequency. If unresponsive to simple treatment, such as increasing fibre in diet, drinking more water or simple laxatives, or if associated with red flag symptoms, then investigation may be needed. Common causes for constipation are shown in Table 14.2 and it may be useful to consider these while taking the history to assess constipation. Any new onset constipation that persists for weeks in a

middle-aged patient is a red flag and requires urgent assessment, although is less concerning than an alteration to looser bowels.

Patients may report being constipated when they sense that they have not adequately emptied their bowel by defaecation, and this is referred to as sense of incomplete evacuation. It is important to identify this symptom because it may be associated with rectal lesions, including rectal cancers, and will require urgent assessment, especially if of recent onset. Tenesmus is pain in the peri-anal area during defaecation or the sense of needing to pass stool when there is none there.

Diarrhoea

Diarrhoea is an increase in the frequency of bowel movements; however, because there is a wide variation in the norm, it is important to establish in the history what was normal for a patient. However, passage of more than three stools per day or the passage of a large amount of stool (>300 g/day) will be considered as diarrhoea. It is important to ask the following key questions while taking a history of diarrhoea to help ascertain the likelihood of underlying diseases:

1. Is it acute (present for days/weeks) or chronic (greater than 3 months to years)?
2. Does it contain blood? Does it contain mucus?
3. Does it wake the patient in the night?
4. Is it accompanied by weight loss?

Acute diarrhoea commonly results from viral or bacterial infection, whereas longer periods of diarrhoea (chronic diarrhoea) may have a variety of causes as shown in Table 14.3. The nature of the stool may give clues as to the underlying cause. In inflammatory bowel disease, the stool will be mixed with blood and mucus. The presence of blood in the stool excludes irritable bowel syndrome as the primary cause of any diarrhoea. In contrast, steatorrhoea is a distinct term that refers to the passage of pale, bulky stools containing excess fats that commonly float in water and are difficult to flush away. It is often described in patients with fat malabsorption owing to chronic pancreatitis with lipase deficiency or small bowel diseases with bile salt malabsorption. It is also important to distinguish diarrhoea from symptoms of incontinence or urgency as the underlying causes may be different.

Table 14.2 Causes of constipation

Types	Causes
Life style related	Low fibre, low fluid intake, drug related
Structural	Carcinoma, strictures, in colon or small bowel
Systemic	Metabolic (hypothyroid), neurological
Idiopathic/dysmotility	Irritable bowel syndrome

Rectal bleeding

Bleeding from the rectum or anal canal typically causes bright red blood loss that is separate from the stool or just noticeable on the toilet paper: haemorrhoids are the most common cause. If darker red and mixed with the stool, this usually indicates a source above the rectum, of which carcinoma is the most important cause. Haemoatochezia is passage of large-volume red blood per rectum and, if not from the colon, may indicate a potentially exsanguinating bleed from an upper GI source.

Flatulence

Flatulence describes passage of excessive wind. It is associated with belching, abdominal distension and passage of flatus per rectum. It is often associated with excessively swallowed air. In some patients it is associated with ingestion of certain foods, such as legumes, beans, broccoli and cabbage. It is a socially embarrassing symptom yet infrequently associated with structural diseases of the GI tract.

Jaundice

Jaundice (or icterus) is a yellowish pigmentation of the skin and conjunctival membranes caused by high levels of bilirubin in the blood, which is a waste product from the breakdown of red blood cells. Its presence implies disease of the liver or the biliary tract, although it may also be the result of excessive haemolysis. Jaundice may be classified depending on its cause as pre-hepatic, owing to excess bilirubin from haemolysis; hepatic, owing to diseases of the liver; or post-hepatic, caused by obstruction in the flow of the bile within the biliary tree owing to

Table 14.3 Causes of diarrhoea

	Causes	Clues
Inflammatory	Infections, ulcerative colitis, Crohn's disease	Blood in stools, abdominal pain, fever
Dysmotility	Irritable bowel syndrome	Abdominal pain Does not contain blood, not interrupt sleep
Malabsorption	Small bowel, pancreatic disease	Steatorrhoea, weight loss, anaemia
Secretory	Villous adenoma, ileal resection	Large volume No change with fasting

Table 14.4 Types of jaundice

	Pre-hepatic	Hepatic	Post-hepatic
History	No itch, Normal colour stools	May have itch Normal/pale stools	Itch Pale-coloured stool
Urine	Normal	Dark	Dark
Sclera	Lemon tinge	Dark	May fluctuate
Serum bilirubin	Often less than 70	Variable	Can go very high Plateaus at 500
Diseases	Gilbert's, drug-induced haemolysis	Hepatitis, cirrhosis, liver cancer	Stones in common bile duct Cancer head of pancreas, biliary tree

diseases of the biliary tree or pancreas. An important aspect of history taking is to be able to differentiate these three types of jaundice. The important questions to ask in the history are:

1. Presence of change in colour of urine
2. Presence of clay-coloured stools
3. Presence of itch

Haemolysis results in excess production of unconjugated bilirubin, which is water insoluble and does not get filtered in the urine. Hence, in jaundice caused by haemolysis, there is no change in colour of urine. In contrast, in hepatic and post-hepatic jaundice (also known as obstructive jaundice), the predominant type of bilirubin in the circulation is conjugated bilirubin which is water soluble and gets filtered out in the urine so the urine colour is darker or deeper yellow. In addition, post-hepatic jaundice is characterized by pale stools caused by a block in the passage of bile into the bowel and itch attributed to reflux of bile salts into the circulation. The differences between the three types of jaundice are shown in Table 14.4. In addition to the characteristic colour of the skin and conjunctiva, there may be other associated cutaneous and systemic features of liver disease, often with dark urine (see below).

Pruritus

Pruritus or itch is a symptom associated with liver disease or obstructive jaundice. It is most often seen in patients with liver diseases, such as primary biliary cholangitis, primary sclerosing cholangitis and cholestasis of pregnancy where it may be the only symptom and not accompanied by jaundice. In contrast, it often accompanies jaundice in patients with carcinoma of the head of the pancreas or a cholangiocarcinoma. Pruritus is attributed to the presence of elevated bile salts and diminishes more quickly than the jaundice when the biliary obstruction is relieved.

Anorexia

Anorexia, which refers to loss of appetite, may be associated with a wide variety of conditions, both GI and non-GI-related. It is important to recognize the symptom of anorexia as distinct from the psychiatric illness *anorexia nervosa*. Another symptom to look for is early satiety where a patient may report feeling hungry but feels full after a few mouthfuls; this is indicative of decreased gastric capacity, either owing to extrinsic compression from an enlarged liver or spleen, an infiltrating type of gastric cancer called 'linitis plastica', which can reduce gastric capacity to distend, or more generally in patients with upper GI malignancy.

Box 14.8 Peripheral stigmata (signs) of chronic liver disease

Skin, nails and hands

- Clubbing of the fingers
- Leuconychia: Expansion of the paler half-moon at the base of the nail
- Palmar erythema: Seen on the thenar and hypothenar eminence, often with blotchy appearance
- Bruising
- Dupuytren's contracture: Can occur in the absence of liver disease
- Hepatic flap: A sign of encephalopathy and advanced disease
- Scratch marks: Particularly in cholestatic liver disease

Face, neck, chest, groins

- Parotid swelling: Particularly in alcohol-related liver disease
- Spider naevi: Small telangiectatic superficial blood vessels with a central feeding vessel
- Hepatic foetor: Characteristic sweet-smelling breath
- Gynaecomastia
- Testicular atrophy: Loss of axillary and pubic hair

Feet

- Pedal oedema

Weight loss

Weight loss may be owing to a decrease in food intake (as seen with anorexia, dysphagia or vomiting), malabsorption of nutrients or the effect of systemic diseases. Diseases that directly cause malabsorption include coeliac disease, pancreatic exocrine insufficiency and, occasionally, inflammatory bowel disease. Weight loss is also commonly associated with systemic conditions, such as cancer (within or outside the GI tract); chronic infections, such

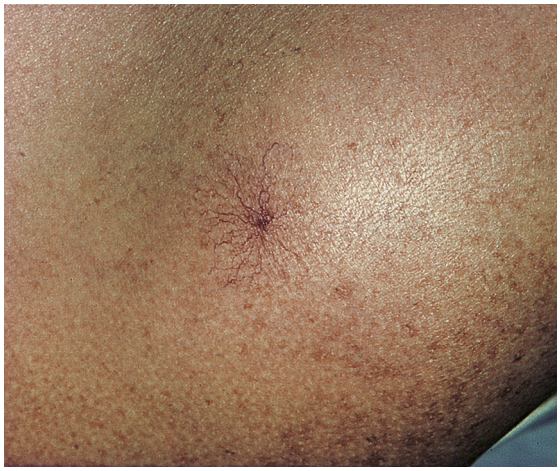


Figure 14.3 A typical spider naevus, with a central arteriole and fine radiating vessels.



Figure 14.4 Palmar erythema in chronic liver disease (sparing the centre of the palms).

as tuberculosis (within or outside the GI tract); and endocrine conditions, such as thyrotoxicosis (despite an increased appetite) and Addison's disease.

Urinary symptoms

Urinary symptoms are discussed in Chapter 17.

Physical examination of the GI tract and abdomen

Physical examination involves a stepwise approach through the four steps of inspection, palpation, percussion and auscultation. In addition, for the GI tract, clusters of signs are linked with specific diseases and it is important to be aware of these as they provide clues to an underlying disease. Signs associated with chronic liver disease are shown in Box 14.8; of these, the most common and useful are spider naevi (Fig. 14.3) (the presence of up to five can be normal) and palmar erythema (Fig. 14.4) (the blotchy appearance often being more important than the overall redness). Inflammatory bowel disease may give rise to clubbing of the hands, arthritis, uveitis and skin changes, including erythema nodosum (tender raised red lumps on the extensor surface of the limbs) and the much rarer pyoderma gangrenosum. Anaemia accompanies many GI diseases, as does oedema, and lymphadenopathy can be secondary to GI malignancy.

It is helpful when examining the patient to record notes or communicate information to colleagues to remember the surface anatomy of the structures related to the GI tract and abdomen (Figs 14.5 and 14.6) and to think of the abdomen as divided into regions (Fig. 14.7). The two lateral vertical planes pass from the femoral artery below to cross the costal margin close to the tip of the ninth costal cartilage. The two horizontal planes, the subcostal and interiliac, pass across the abdomen to connect the lowest points on the costal margin and the tubercles of the iliac crests, respectively.

Beginning the abdominal examination

1. Wash your hands and introduce yourself (if you examining as a student on a ward round; if you took the history, you will have already introduced yourself).
2. Confirm the patient's identity and explain the different steps of the abdominal examination in clear and easy to understand language, to seek patient agreement and consent for proceeding with the examination.
3. Position the patient appropriate for the examination: The patient should be lying supine with arms loosely at the sides, the head and neck supported by up to two pillows, sufficient for comfort. Make sure there is a good light, that the room is warm and that the hands are warm. A shivering patient cannot relax and vital signs, especially on palpation, may be missed.
4. Exposure of the abdomen: although it is important for the purpose of this examination that the area from 'nipples to knees' should be examined, it is not acceptable or appropriate to expose this area all at once. Instead, a stepwise approach is best, beginning with asking the patient to draw clothing up to just above the xiphisternum and a sheet folded down to the level of the symphysis pubis. The examination of the groins and genitalia must not be neglected and needs to be carried out with discretion, with full explanation as to the reasons, and leaving these areas exposed for a minimum amount of time. It is not unusual for a patient to present with intestinal obstruction owing to a strangulated femoral or inguinal hernia where the diagnosis has been missed initially owing to lack of proper inspection of the groins in an effort to save embarrassment.

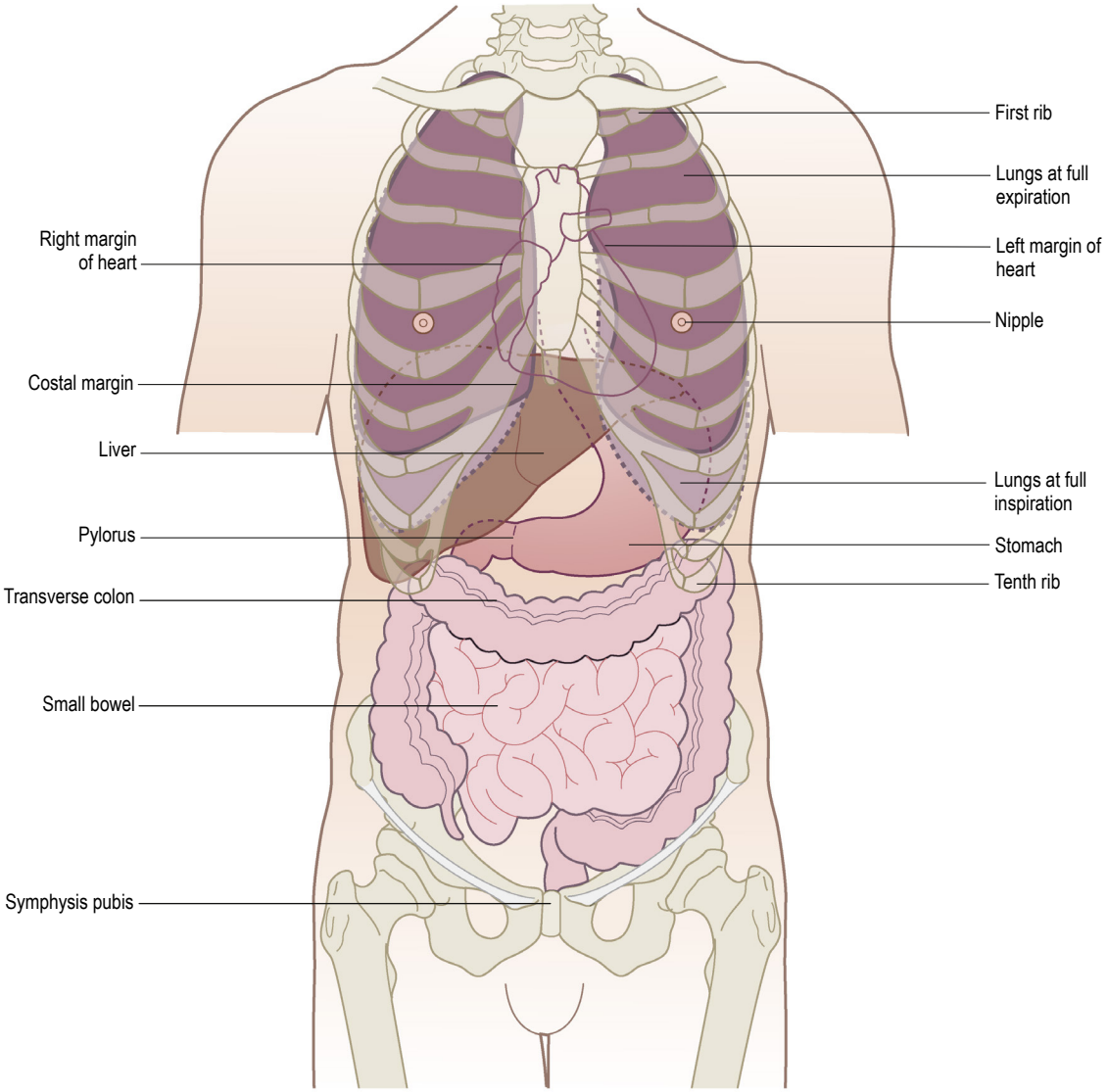


Figure 14.5 Anterior view of the external relationships of the abdominal and thoracic organs.

5. The last thing before beginning the examination is to ask if the patient is in pain and if he is comfortable to proceed.

Inspection

Inspection is an important and neglected part of abdominal examination. Initially, it is well worthwhile spending 30 seconds observing the abdomen from different positions to note the following features:

Shape

Is the abdomen of normal contour and fullness, or distended? Is it scaphoid (sunken)?

- Generalized fullness or distension may be caused by fat, fluid, flatus, faeces or foetus.
- Localized distension may be symmetrical and centred around the umbilicus as in the case of small bowel obstruction, or asymmetrical as in gross enlargement of the spleen, liver or ovary.
- Make a mental note of the site of any such swelling or distension; think of the anatomical structures in that region and note if there is any movement of the swelling, either with, or independent of, respiration.
- Remember that chronic urinary retention may cause palpable enlargement of the bladder felt in the lower abdomen.
- A scaphoid abdomen is seen in advanced stages of starvation and malignant disease.

The umbilicus

Normally the umbilicus is slightly retracted and inverted. If it is everted, then an umbilical hernia may be present; this can be confirmed by feeling an expansile impulse on palpation of the swelling when the patient coughs. The hernial sac may contain omentum, bowel or fluid. A common finding in the umbilicus of elderly obese people is a concentration

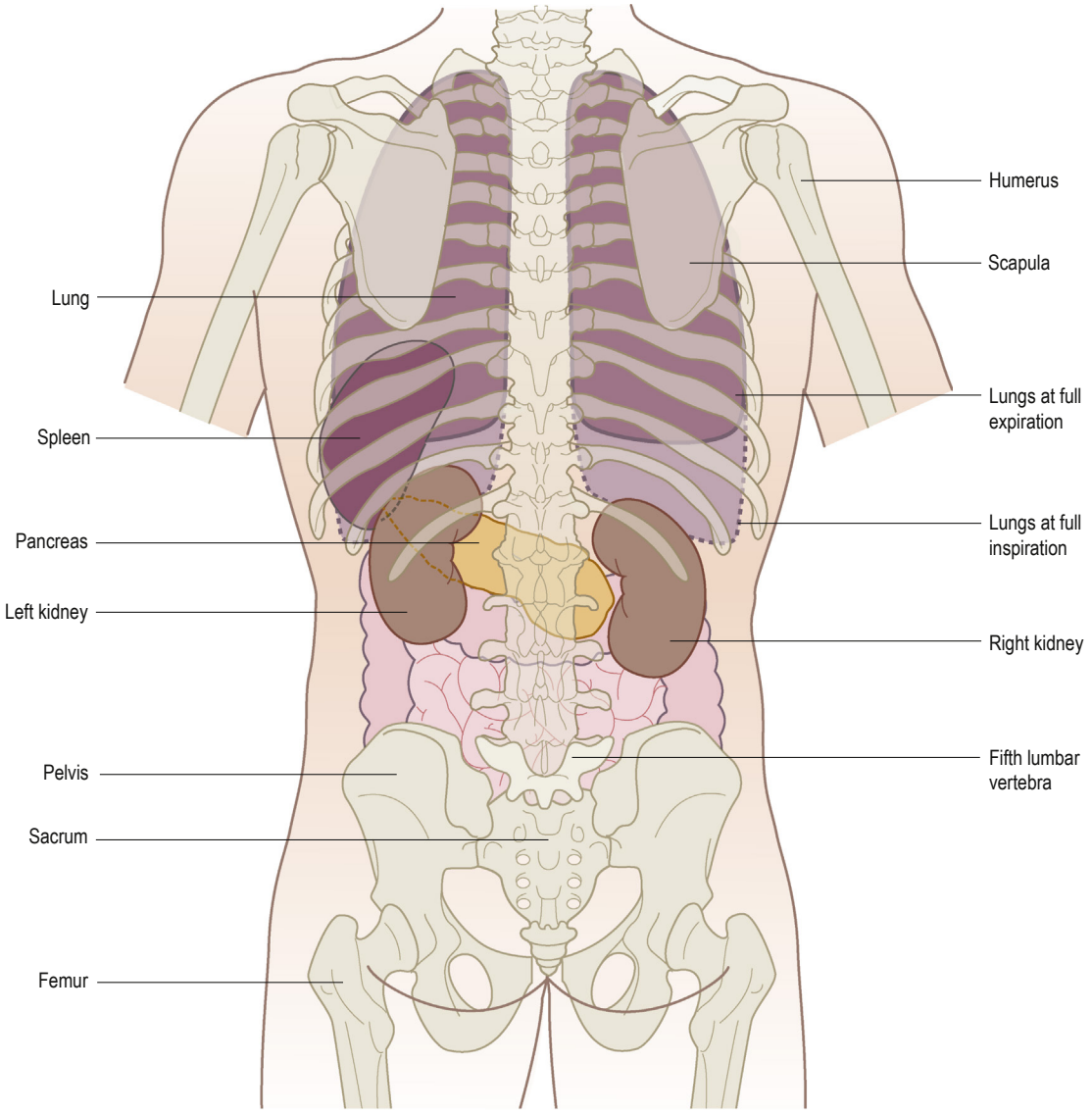


Figure 14.6 Posterior view of the external relationships of the abdominal and thoracic organs. The liver is not shown.

of inspissated desquamated epithelium and other debris (omphalolith).

Skin and surface of the abdomen

On inspection of the surface of the abdomen, look for the following characteristics: scars, striae, superficial veins, stomas and pigmentation.

1. Scars: Note any scars present, their site, whether they are old (white) or recent (red or pink), linear or stretched (and therefore likely to be weak and contain an incisional hernia). Common examples of post-surgery scars are given in Figure 14.8.
2. Striae: Striae atrophica or gravidarum are white or pink wrinkled linear marks on the abdominal skin. Produced by gross stretching of the skin with rupture of the elastic fibres, they indicate a recent change in size of the abdomen, such as is found in pregnancy, ascites, wasting diseases and severe dieting. Wide purple striae are characteristic of Cushing's syndrome.
3. Superficial veins: Look for prominent superficial veins, which may be apparent in three situations (Fig. 14.9: thin veins over the costal margin, usually of no significance; occlusion of the inferior vena cava; and venous anastomoses in portal hypertension). Obstruction of the inferior vena cava not only causes oedema of the limbs, buttocks and groins but, in time, distended veins on the abdominal wall and chest wall appear. These represent dilated anastomotic channels between the superficial epigastric and circumflex iliac veins below and the lateral thoracic veins above, conveying the diverted blood from the long saphenous vein to the axillary vein; the direction of flow is therefore upwards. If the

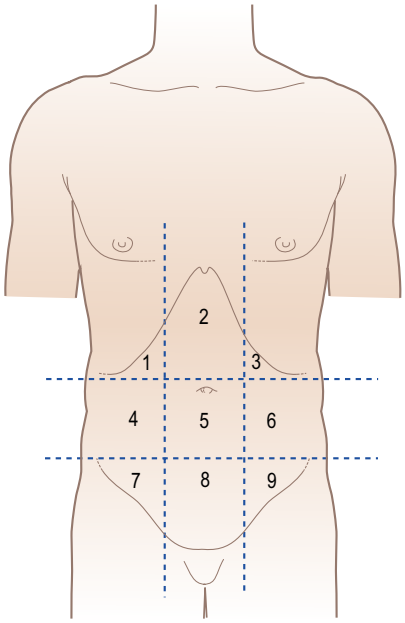


Figure 14.7 Regions of the abdomen. 1 and 3, right and left hypochondrium; 2, epigastrium; 4 and 6, right and left lumbar; 5, umbilical; 7 and 9, right and left iliac; 8, hypogastrium or suprapubic.

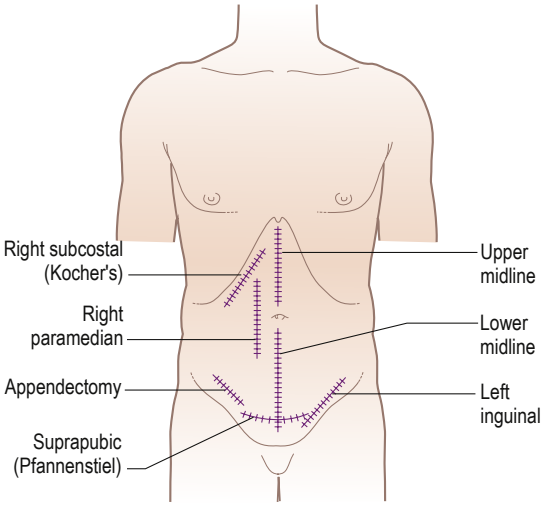


Figure 14.8 Some commonly used abdominal incisions. The midline and oblique incisions avoid damage to the innervation of the abdominal musculature and the later development of incisional hernia.

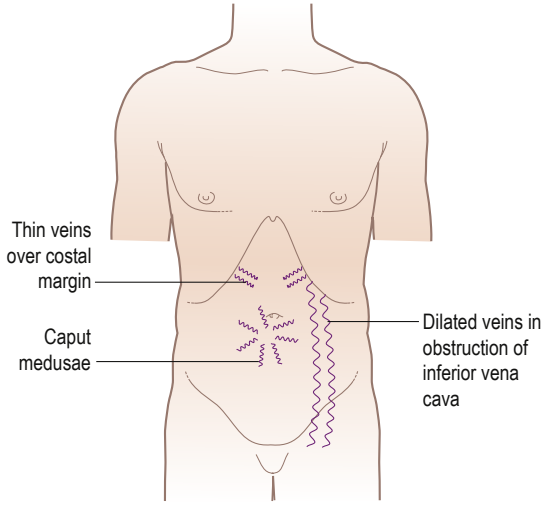


Figure 14.9 Prominent veins of the abdominal wall.

veins are sufficiently prominent, try to detect the direction in which the blood is flowing by occluding a vein, emptying it by massage and then looking for the direction of refill. Distended veins around the umbilicus (caput medusae) are uncommon, but signify portal hypertension, other signs of which may include splenomegaly, ascites and other cutaneous manifestations of chronic liver disease. These distended veins represent the opening up of anastomoses between portal and systemic veins and occur in other sites, such as oesophageal and rectal varices.

4. Stomas: Stomas are surgically created openings in the abdominal wall and are identified by the presence of a stoma bag overlying these. There are three different types of stoma: a colostomy, an ileostomy or a urosotomy (Fig. 14.10). The ileostomy, often on the right iliac fossa, is identified by an elevated spout. Colostomy stomas are often flat and can be either an end colostomy or a loop colostomy, in which case two lumens may be seen within the same stoma site. Urostomy is used to divert urine into a loop of bowel in the event of resection of urinary bladder.
5. Pigmentation: Pigmentation of the abdominal wall may be seen in the midline below the umbilicus, where it forms the *linea nigra* and is a sign of pregnancy. Erythema *ab igne* is a brown mottled pigmentation produced by constant application of heat, usually a hot water bottle or heat pad, on the skin of the abdominal wall or back. It is a sign that the patient is experiencing severe persistent pain, such as from chronic pancreatitis.

Palpation

Palpation forms the most important part of the abdominal examination. Ask the patient to relax as much as possible, to breathe quietly and assure that you will be as gentle as possible. Enquire about the site of any pain and examine this region last. These points, together with unhurried palpation with a warm hand, will give the patient confidence and allow the maximum amount of information to be obtained.

When palpating, the wrist and forearm should be in the same horizontal plane where possible, even if this means bending down or kneeling by the patient's side. The best palpation technique involves moulding the relaxed right hand to the abdominal

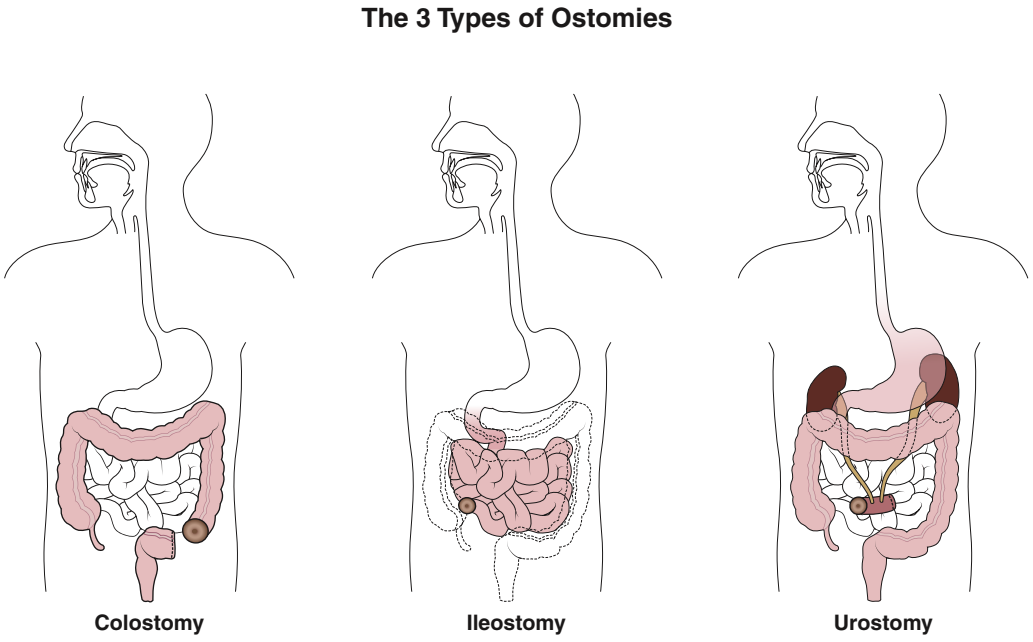


Figure 14.10 Types of stomas on abdominal examination.



Figure 14.11 Correct method of palpation. The hand is held flat and relaxed and 'moulded' to the abdominal wall.

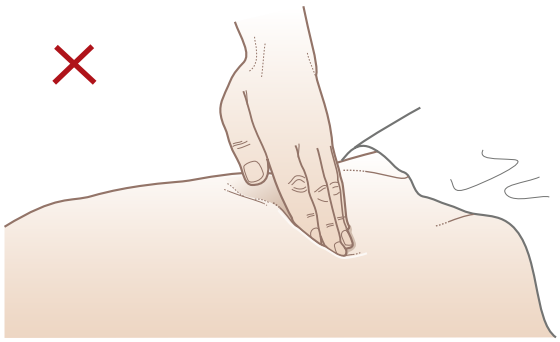


Figure 14.12 Incorrect method of palpation. The hand is held rigid and mostly not in contact with the abdominal wall.

wall, not to hold it rigid (Fig. 14.11). The best movement is gentle but with firm pressure, with the fingers held almost straight but with slight flexion at the metacarpophalangeal joints and avoiding sudden poking with the fingertips (Fig. 14.12).

Palpation of intra-abdominal structures is an imperfect process in which the great sensitivity of the sense of touch and pressure is heavily masked by the abdominal wall tissue. It is unusual for structures to be very easily palpable and so it is necessary to concentrate fully on the task and to try to visualize the normal anatomical structures and what might be palpable beneath the examining hand. It may be necessary to repeat the palpation more slowly and deeply. Putting the left hand on top of the right allows increased pressure to be exerted (Fig. 14.13), such as with an obese or very muscular patient.

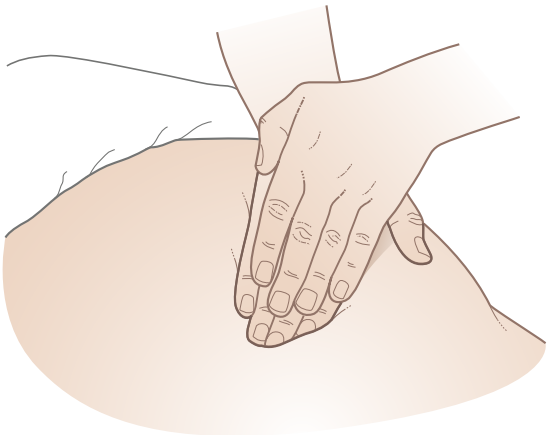


Figure 14.13 Method of deep palpation in an obese, muscular or poorly relaxed patient.

A small proportion of patients find it impossible to relax their abdominal muscles when being examined. In such cases, it may help to ask them to breathe deeply, to bend their knees up or to distract their attention in other ways. No matter how experienced the examiner, little will be gained from palpation of a poorly relaxed abdomen. It is important to keep in mind the following three signs on palpation which can be markers of serious intra-abdominal pathology.

Guarding

Guarding is an involuntary reflex contraction of the muscles of the abdominal wall overlying an inflamed viscus and peritoneum, producing localized rigidity. It indicates localized peritonitis. What is felt on examination is spasm of the muscle, which prevents palpation of the underlying viscus. Guarding is seen classically in uncomplicated acute appendicitis. It is very important to distinguish this sign from the voluntary contraction of muscle.

Rigidity

Generalized or 'board-like' rigidity is an indication of diffuse peritonitis. It can be looked upon as an extension of guarding, with involuntary reflex rigidity of the muscles of the anterior abdominal wall. It is quite unmistakable on palpation; the whole abdominal wall feels hard and 'board-like', precluding palpation of any underlying viscus. The least downward pressure with a palpating hand in a patient with generalized rigidity produces severe pain. It may be differentiated from voluntary spasm by getting the patient to breathe: if there is voluntary spasm, the abdominal wall will be felt to relax during expiration.

Rebound tenderness

Rebound tenderness is present if, when palpating slowly and deeply over a viscus and then suddenly releasing the palpating hand, the patient experiences sudden pain. Rebound tenderness is not always a reliable sign and should be interpreted with caution, particularly in those patients with a low pain threshold, but is often a useful adjunct to detecting peritoneal inflammation.

To ensure comprehensive assessment of all intra-abdominal structures, it is helpful to have a logical sequence to follow and, if this is done as a matter of routine, then no important point will be omitted. Always consider the underlying anatomy when examining the abdomen. The following scheme is suggested, which may need to be varied according to the site of any pain, palpating the site of pain towards the end:

1. Light palpation: Start in the left lower quadrant of the abdomen, palpating lightly, and move in an anti-clockwise direction including all nine areas of the abdomen; adopting this routine reduces the risk of missing signs.
2. Deep palpation: Repeat using slightly deeper palpation examining each of the nine areas of the abdomen.
3. Focused palpation:
   - **a.** Feel for the liver and gall bladder.
   - **b.** Feel for the spleen.
   - **c.** Feel for the left kidney.
   - **d.** Feel for the right kidney.
   - **e.** Feel for the urinary bladder.
   - **f.** Feel for the aorta and para-aortic glands and common femoral vessels.
   - **g.** Palpate both groins.
   - **h.** Examine the external genitalia.
   - **i.** If a swelling is palpable, spend time eliciting its features.

All of the organs in the upper abdomen (liver, spleen, kidneys, stomach, pancreas, gallbladder) move downwards with inspiration (with the spleen moving more downwards and medially). Thus, asking the patient to take a deep breath while examining makes detection of these organs easier. When the patient breathes in, the examining hand should be still so that the organ in question 'comes on to the examining hand' or 'slips by underneath it'.

Liver

24th edition this is the third organ

Place the right hand below and parallel to the right subcostal margin. The liver edge will then be felt against the radial border of the index finger (Fig. 14.14). The liver is often palpable in normal patients without being enlarged. The lower edge of the liver can be clarified by percussion (see below), as can the upper border in order to determine overall size: a palpable liver edge can be owing to enlargement, or displacement downwards by lung pathology. Hepatomegaly conventionally is measured from the edge of the right costal margin in the mid-clavicular line. The degree of liver enlargement is often quoted in finger breadths, but

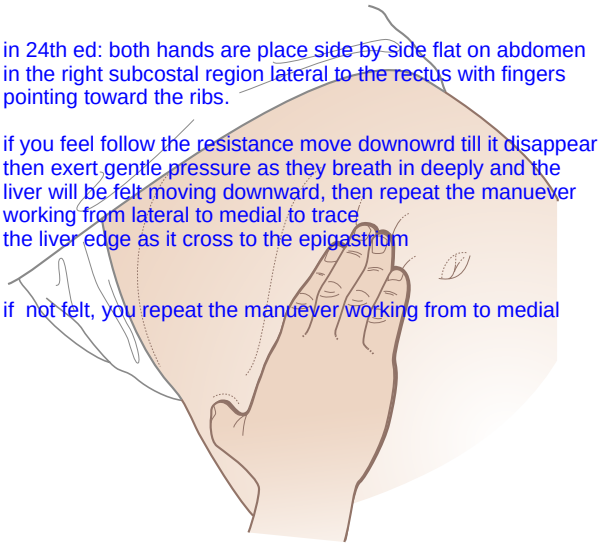


Figure 14.14 Palpation of the liver.

these are variable and it is better to use centimeters. One helpful tip is to place the left hand over the right costal margin and use the right hand to assess for hepatomegaly, starting from the right iliac fossa and moving the right hand gradually upwards with each inspiration.

Try to discern the character of the liver surface (i.e. whether it is soft, smooth and tender as in heart failure, very firm and regular as in obstructive jaundice and cirrhosis, or hard, irregular, painless and sometimes nodular as in advanced secondary carcinoma). In tricuspid regurgitation, the liver may be felt to pulsate. Occasionally a congenital variant of the right lobe, called Riedel's lobe, projects down lateral to the gallbladder as a tongue-shaped process. Although uncommon, it is important to be aware of this because it may be mistaken either for the gallbladder itself or for the right kidney.

Gallbladder

The gallbladder is palpated in the same way as the liver, although the normal gallbladder usually is not palpable. When it is distended, however, it forms an important sign and may be palpated as a firm, smooth or globular swelling with distinct borders, just lateral to the edge of the rectus abdominis near the tip of the ninth costal cartilage. It moves with respiration. Its upper border merges with the lower border of the right lobe of the liver or disappears beneath the costal margin and therefore can never be felt (Fig. 14.15). When the liver is enlarged or the gallbladder grossly distended, the latter may be felt not in the hypochondrium but in the right lumbar or even as low down as the right iliac region; hence, the importance of commencing palpation in the right iliac fossa and moving the examining hand upwards. An enlarged gallbladder may be easier to see moving on inspiration than to feel.

The ease of definition of the rounded borders of the gallbladder, its comparative mobility on respiration, the fact that it is not normally bimanually palpable and that it seems to lie just beneath the abdominal wall help to identify such a swelling as the gallbladder rather than as a palpable right kidney. A painless gallbladder usually can be palpated in the following clinical situations:

- In a jaundiced patient with carcinoma of the head of the pancreas or other malignant causes of obstruction of the common bile duct (below the entry of the cystic duct), the ducts above the obstruction become dilated, as does the gallbladder (see Courvoisier's law below).
- In mucocele of the gallbladder, a gallstone becomes impacted in the neck of an uninfected gallbladder and mucus continues to be secreted into its lumen (Fig. 14.16). Eventually, the uninfected gallbladder is so distended that it becomes palpable. In this case, the bile ducts are normal, and the patient is not jaundiced.
- In carcinoma of the gallbladder, the gallbladder may be felt as a stony, hard, irregular swelling, unlike the firm, regular swelling in the two above-mentioned conditions.

Murphy's sign

In acute inflammation of the gallbladder (acute cholecystitis), severe pain is present. Often an exquisitely tender but indefinite mass can be palpated; this represents the underlying acutely inflamed gallbladder walled off by greater omentum. Ask the patient to breathe in deeply and palpate for the gallbladder in the normal way; at the height of inspiration, the breathing stops with a gasp as the mass is felt. This represents Murphy's sign. The sign is not found in chronic cholecystitis or uncomplicated cases of gallstones.

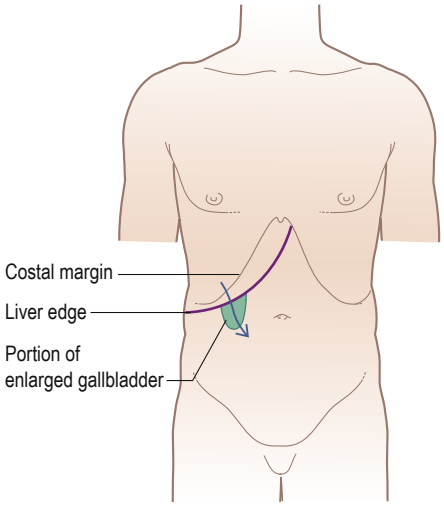


Figure 14.15 Palpation of an enlarged gallbladder, showing how it merges with the inferior border of the liver so that only the fundus of the gallbladder and part of its body can be palpated.



Figure 14.16 A mucocele of the gallbladder that is distended, pale and thin walled.

Courvoisier's law

Courvoisier's law states that in the presence of jaundice, a palpable gallbladder makes gallstone obstruction of the common bile duct an unlikely cause (because it is likely that the patient will have had gallbladder stones for some time and these will have rendered the wall of the gallbladder relatively fibrotic and therefore non-distensible). However, the converse is not true, because the gallbladder is not palpable in many patients who do turn out to have malignant bile duct obstruction.

2nd to palpate per 24th ed

Spleen

Like the left kidney, the spleen is not normally palpable. It has to be enlarged to two or three times its usual size before it becomes palpable and then is felt beneath the left subcostal margin. Enlargement takes place in a superior and posterior direction before it becomes palpable subcostally. Once the spleen has become palpable, the direction of further enlargement is downwards and towards the right iliac fossa (Fig. 14.17). Place the flat of the left hand over the lower-most rib cage posterolaterally and exert gentle traction, thus restricting the expansion of the left lower ribs on inspiration and concentrating more of the inspiratory movement into moving the spleen downwards. The right hand is placed beneath the costal margin more medially. Press in deeply with the fingers of the right hand beneath the costal margin, at the same time exerting considerable pressure medially and downwards with the left hand (Fig. 14.18), and then ask the patient to breathe in deeply. Repeat this manoeuvre with the right hand being moved more towards the left costal margin on each occasion (Fig. 14.19). If enlargement of the spleen is suspected from the history and it is still not palpable, turn the patient half on to the right side, ask him to relax back on to your left hand, which is now supporting the lower ribs, and repeat the examination as above. Alternatively, the spleen may be very large and the lower edge may be much lower than at first suspected.

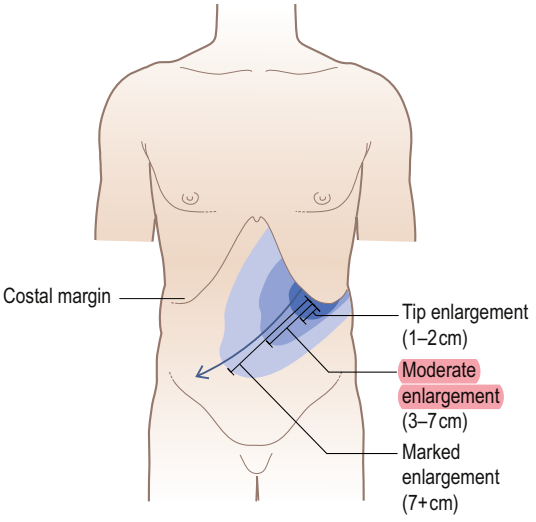


Figure 14.17 The direction of enlargement of the spleen. The spleen has a characteristic notched shape and the organ moves downwards during full inspiration.

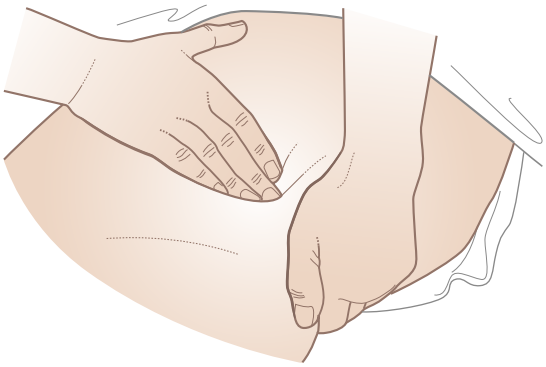


Figure 14.18 Palpation of the spleen. Start well out to the left.

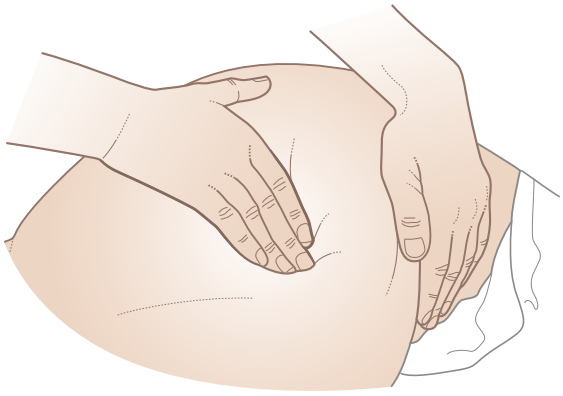


Figure 14.19 Palpation of the spleen more medially than in Figure 14.17.

In minor degrees of enlargement, the spleen will be felt as a firm swelling with smooth, rounded borders. Where considerable splenomegaly is present, its typical characteristics include a firm swelling appearing beneath the left subcostal margin in the left upper quadrant of the abdomen, which is dull to percussion, moves downwards on inspiration, is not bimanually palpable, whose upper border cannot be felt (i.e. one cannot 'get above it') and in which a notch can often, though not invariably, be felt in the lower medial border. The last three features distinguish the enlarged spleen from an enlarged kidney; in addition, there is usually a band of colonic resonance anterior to an enlarged kidney (Table 14.5).

Left kidney

The right hand is placed anteriorly in the left lumbar region while the left hand is placed posteriorly in the left loin (Fig. 14.20). Ask the patient to inhale deeply and press the left hand forward and lift the right hand upward and inward. The left kidney is not usually palpable unless either low in position or enlarged. Its lower pole, when palpable, is felt

Table 14.5 Features to differentiate kidney and spleen on examination

Spleen	Kidney
Dull to percussion	Band of colonic resonance anterior to an enlarged kidney
Moves diagonally downwards towards right iliac fossa on inspiration	Moves vertically downwards inspiration
Not bimanually palpable	Bimanually palpable
Upper border cannot be felt (i.e. one cannot 'get above it')	Can 'get above it'
Notch can often, though not invariably, be felt in the lower medial border	Is not notched

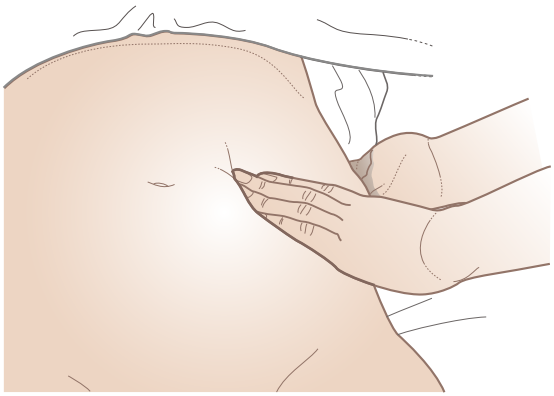


Figure 14.20 Palpation of the left kidney.

as a rounded firm swelling between both right and left hands (i.e. bimanually palpable) and it can be pushed from one hand to the other, in an action which is called 'ballotting'.

Right kidney

Feel for the right kidney in much the same way as for the left. Place the right hand horizontally in the right lumbar region anteriorly with the left hand placed posteriorly in the right loin. Push forwards with the left hand, lift the right hand inward and upward (Fig. 14.21) and ask the patient to inhale deeply. The lower pole of the right kidney, unlike the left, is commonly palpable in thin patients; it is felt as a smooth, rounded swelling which descends on inspiration and is bimanually palpable and may be 'ballotted' (bounced back and forth between the two examining hands).

Urinary bladder

Normally the urinary bladder is not palpable. When it is full and the patient cannot empty it (retention of urine), a smooth, firm, regular oval-shaped swelling

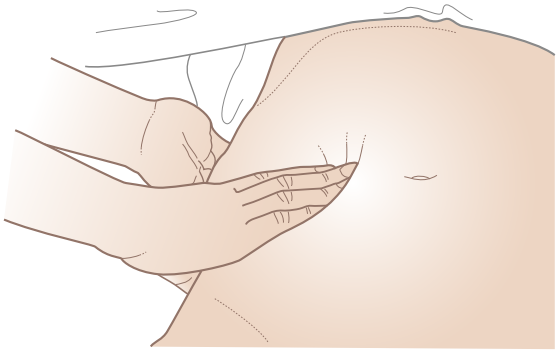


Figure 14.21 Palpation of the right kidney.

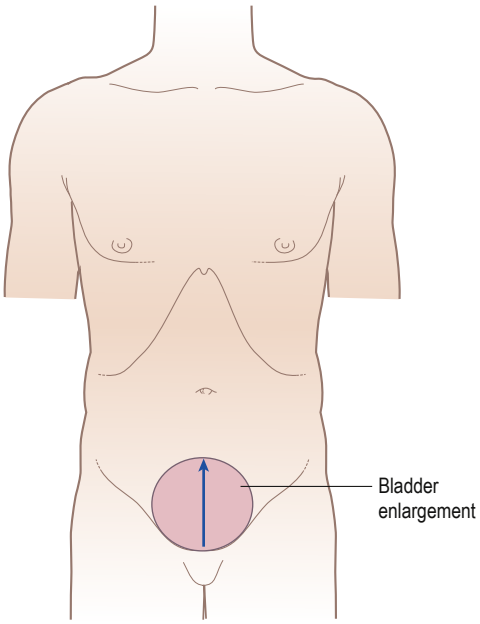


Figure 14.22 Physical signs of urine retention: a smooth, firm and regular swelling arising out of the pelvis which one cannot 'get below' and which is dull to percussion.

will be palpated in the suprapubic region and its dome (upper border) may reach as far as the umbilicus. The lateral and upper borders can be readily identified, but it is not possible to feel its lower border (i.e. the swelling is 'arising out of the pelvis'). The fact that this swelling is symmetrically placed in the suprapubic region beneath the umbilicus, that it is dull to percussion and that pressure on it gives the patient a desire to micturate, together with the signs above, confirms such a swelling as the bladder (Fig. 14.22).

In women, however, a mass that is thought to be a palpable bladder has to be differentiated from a gravid uterus (firmer, mobile side to side), a fibroid uterus (may be bosselated and firmer) and an ovarian cyst (usually eccentrically placed to the left or right side).

The aorta and common femoral vessels

In most adults, the aorta is not readily felt; however, with practice, it usually can be detected by deep



Figure 14.23 Palpation of the abdominal aorta.

palpation a little above and to the left of the umbilicus. In thin patients, particularly women with a marked lumbar lordosis, the aorta is more easily palpable. Palpation of the aorta is one of the few occasions in the abdomen when the fingertips are used as a means of palpation. Press the extended fingers of both hands, held side by side, deeply into the abdominal wall in the position shown in Figure 14.23; identify the left wall of the aorta and note its pulsation. Remove both hands and repeat the manoeuvre a few centimetres to the right. In this way the pulsation and width of the aorta can be estimated. It is difficult to detect small aortic aneurysms; where a large one is present, its presence and width may be assessed by placing the extended fingertips on either side of it with the palms flat on the abdominal wall and the fingers pointing towards each other. When the fingertips are either side of an aneurysm, it should be clear that they are being separated by each pulsation and not just moved up and down (this latter manoeuvre can involve very deep palpation and the patient should be warned). In a patient with known aortic aneurysm or in those at high risk (elderly patients), caution must be exercised with deep palpation because examination can be very uncomfortable for patients and there is a risk of triggering rupture of the aneurysm.

Lymph nodes lying along the aorta (para-aortic nodes) are palpable only when considerably enlarged. They are felt as rounded, firm, often confluent fixed masses in the umbilical region and epigastrium along the left border of the aorta. Pulsations of the aorta are transmitted through the nodes which separates them from the expansile pulsations palpated in aneurysmal dilatation.

Causes of diagnostic difficulty on palpation

In many patients, especially those with a thin or lax abdominal wall, faeces in the colon may simulate an abdominal mass. The sigmoid colon is frequently palpable, particularly when loaded with hard faeces. It is felt as a firm, tubular structure about 12 cm in length, situated low down in the left iliac fossa, parallel to the inguinal ligament. The caecum is often palpable in the right iliac fossa as a soft, rounded swelling with indistinct borders. The transverse colon is sometimes palpable in the epigastrium. It feels somewhat like the sigmoid colon but rather larger and softer, with distinct upper and lower borders and a convex anterior surface. A faecal 'mass' usually will have disappeared or moved on repeat examination and may retain an indentation with pressure (not the case with a colonic malignancy).

In the epigastrium, the muscular bellies of rectus abdominis lying between its tendinous intersections can mimic an underlying mass and give rise to confusion. This usually can be resolved by asking the patient to tense the abdominal wall (by lifting the head off the pillow), when the 'mass' may be felt to contract.

What to do when an abdominal mass is palpable

An abdominal mass is an important finding and requires further characterization in order to establish the organ of origin and its pathological nature. The following are key parameters to include in the examination of a palpable abdominal mass.

Site

Note the region occupied by the swelling. Think of the organs that normally lie in or near this region and consider whether the swelling could arise from one of these organs. For instance, a swelling in the right upper quadrant most probably arises from the liver, right kidney, hepatic flexure of colon or gallbladder.

Feeling the swelling while the patient lifts his head and shoulders off the pillow to tense the anterior abdominal wall will differentiate between a mass in the abdominal wall and within the abdominal cavity. If the swelling is in the upper abdomen, try to determine if it is possible to 'get above it'; that is, to feel the upper border of the swelling as it disappears above the costal margin, and similarly, if it is in the lower abdomen, whether one can 'get below it'. If one cannot 'get above' an upper abdominal swelling, a hepatic, splenic, renal or gastric origin should be suspected. If one cannot 'get below' a lower abdominal mass, the swelling probably arises in the bladder, uterus, ovary or, occasionally, upper rectum.

Size and shape

It is important to characterize the outline of the lump and whether it is symmetrical or asymmetrical in nature. The larger a swelling arising from one of these structures, the more it tends to distort the outline of the organ of origin (e.g. a large renal mass can feel as if it is arising from intraperitoneal organs). Second, establish an approximate diameter of the lump, ideally using a tape measure.

Surface, edge and consistency

The pathological nature of a mass is suggested by a number of features. A swelling that is hard, irregular in outline and nodular is likely to be malignant,

whereas a regular, round, smooth, tense swelling is likely to be cystic. A solid, ill-defined and tender mass suggests an inflammatory lesion such as may be seen in ileocaecal Crohn's disease.

Temperature and tenderness

Examine the overlying skin with the dorsum of the hand, comparing the temperature over the skin overlying the lump with that of the rest of abdomen. Inspect whether there are any overlying changes in the skin, such as erythema or tense skin overlying the lump. Elicit whether the lump is tender to palpation. Hot, tender lumps are likely to be inflammatory in origin or a site of infection.

Mobility and attachments

Swellings arising in the liver, spleen, kidneys, gallbladder and distal stomach all show downward movement during inspiration, owing to the normal downward diaphragmatic movement, and such structures cannot be moved with the examining hand. In contrast, tumours of the small bowel and transverse colon, cysts in the mesentery and large secondary deposits in the greater omentum are not usually influenced by respiratory movements, but may easily move on palpation.

When the swelling is completely fixed, it usually signifies one of three things:

1. A mass of retroperitoneal origin (e.g. pancreas)
2. Part of an advanced tumour with extensive spread to the anterior or posterior abdominal walls or abdominal organs
3. A swelling resulting from severe chronic inflammation involving other organs (e.g. diverticulitis of the sigmoid colon or a tuberculous ileocaecal mass)

In the lower abdomen, the side-to-side mobility of a fibroid or pregnant uterus rapidly establishes such a swelling as uterine in origin and as not arising from the urinary bladder.

Is it bimanually palpable or pulsatile?

Bimanually palpable swellings in the lumbar region are usually renal in origin. Occasionally, however, a posteriorly situated gallbladder or a mass in the postero-inferior part of the right lobe of the liver may give the impression of being bimanually palpable. Carefully note whether a swelling is pulsatile and decide if any pulsation comes from the mass or is transmitted through it.

Percussion

Details of how to percuss correctly are given in Chapter 12. The normal percussion note over most of the abdomen is resonant (tympanic) except over the liver, where the note is dull. A normal spleen is not large enough to render the percussion note dull. A resonant percussion notes over suspected enlargement of liver or spleen weighs against there being true enlargement.

In obese patients, tympanic areas of the abdomen may not give a truly resonant percussion note and palpation of organs such as a large liver is more difficult. If hepatomegaly is suspected, rhythmic percussion just above the suspected lower border of the liver, as the patient breathes in and out deeply, can elicit a note cyclically changing between dull to hollow, and eliciting this change may be more certain than the character of the fixed and unchanging note.

Defining the boundaries of abdominal organs and masses

Liver

The upper and lower borders of the right lobe of the liver can be mapped out accurately by percussion. Start anteriorly, at the fourth intercostal space, where the note will be resonant over the lungs, and work vertically downwards.

Over a normal liver, percussion will detect the upper border, which is found at about the fifth intercostal space (just below the right nipple in men). The dullness extends down to the lower border at or just below the right subcostal margin, giving a normal liver vertical height of 12–15 cm. The normal dullness over the upper part of the liver is reduced in severe emphysema, in the presence of a large right pneumothorax and after laparotomy or laparoscopy.

Spleen

Percussion over a substantially enlarged spleen provides rapid confirmation of the findings detected on palpation (see Fig. 14.18). Dullness extends from the left lower ribs into the left hypochondrium and left lumbar region.

Urinary bladder

The findings in a patient with retention of urine are usually unmistakable on palpation (see Fig. 14.22). The dullness on percussion and clear difference from the adjacent bowel provides reassurance that the swelling is cystic or solid and not gaseous.

Other masses

The boundaries of any localized swelling in the abdominal cavity or in the walls of the abdomen sometimes can be defined more accurately by percussion than palpation, as for the urinary bladder.

Detection of ascites and its differentiation from ovarian cyst and intestinal obstruction

There are three common causes of diffuse enlargement of the abdomen:

1. The presence of free fluid in the peritoneum (ascites)
2. A massive ovarian cyst
3. Obstruction of the large bowel, distal small bowel or both

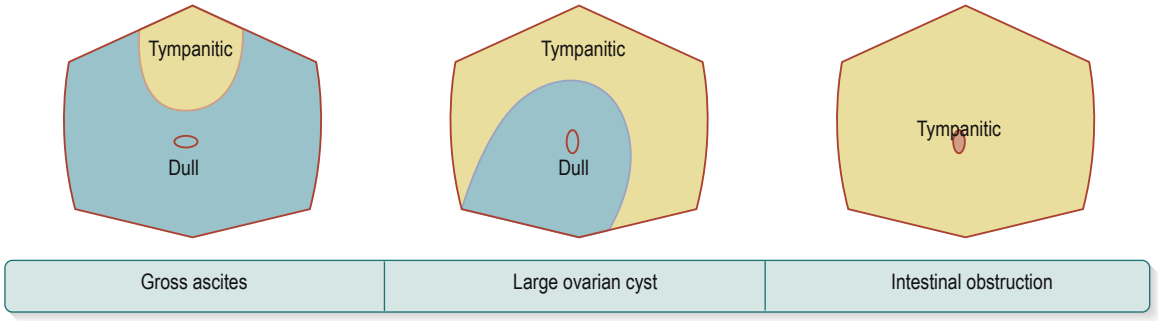


Figure 14.24 Three types of diffuse enlargement of the abdomen.

Box 14.9 Clinical features of marked abdominal swelling

Gross ascites

- Dull in flanks
- Umbilicus everted and/or hernia present
- Shifting dullness positive
- Fluid thrill positive

Large ovarian cyst

- Resonant in flank
- Umbilicus vertical and drawn up
- Large swelling felt arising out of pelvis which one cannot 'get below'

Intestinal obstruction

- Resonant throughout
- Colicky pain
- Vomiting
- Recent cessation of passage of stool and flatus
- Increased and/or 'tinkling' bowel sounds

Percussion rapidly distinguishes between these three causes, as can be seen in Figure 14.24. Other helpful symptoms or signs which are usually present are listed in Box 14.9.

The use of ultrasound to assess ascites has shown that at least 2 litres of ascites needs to be present to be detected clinically. It is unreliable to diagnose ascites unless sufficient free fluid is present to give generalized enlargement of the abdomen. The cardinal sign created by ascites is shifting dullness. A fluid thrill may also be present, but it would be unwise to diagnose ascites based on this sign without the presence of shifting dullness.

To demonstrate shifting dullness, ask the patient to lie supine. Place your fingers in the longitudinal axis on the midline near the umbilicus and begin percussion, moving your fingers laterally towards the right flank (Fig. 14.25). When dullness is first detected (in normal individuals, dullness is only over the lateral abdominal musculature), keep your fingers in that position and ask the patient to roll on to his left side. Wait a few seconds for any peritoneal fluid to redistribute and, if ascites is present, the percussion note will become resonant. This shift in the area of dullness can be confirmed by finding the left border of dullness with the patient still on his left side and seeing if it shifts when the patient returns to the supine position or by repeating the original manoeuvre but towards the other side of the abdomen. It is important to have assessed for an enlarged liver or spleen before eliciting shifting dullness and, if present, ask the patient to roll over to the opposite side when eliciting shifting dullness.

To elicit a fluid thrill, the patient again lies supine. Place one hand flat over the lumbar region of one side, and get an assistant to put the side of his hand longitudinally and firmly in the midline of the abdomen. Then flick or tap the opposite lumbar region (Fig. 14.26). A fluid thrill or wave is felt as a definite and unmistakable impulse by the detecting hand held flat in the lumbar region. (The purpose of the assistant's hand is to dampen any impulse that may be transmitted through the fat of the abdominal wall.) As a rule, a fluid thrill is felt only when a large amount of ascites is present, which is under tension, and it is not a very reliable sign.

Auscultation

Auscultation of the abdomen is done to detect bowel sounds and vascular bruits.

Bowel sounds

The stethoscope should be placed on one site on the abdominal wall (just to the right of and below the umbilicus is best) and kept there until sounds are heard. It should not be moved from site to site. Normal bowel sounds are heard as intermittent low- or medium-pitched gurgles interspersed with an occasional high-pitched noise or tinkle.

In simple acute mechanical obstruction of the small bowel, the bowel sounds are excessive and exaggerated. Frequent, loud, low-pitched gurgles (borborygmi) are heard, often with a crescendo to high-pitched tinkles and occurring in a rhythmic pattern with peristaltic activity. The presence of such sounds occurring at the same time as the patient experiences bouts of colicky abdominal pain is highly suggestive of small bowel obstruction. In between the bouts of peristaltic activity and colicky pain, the bowel is quiet and no sounds are heard on auscultation.

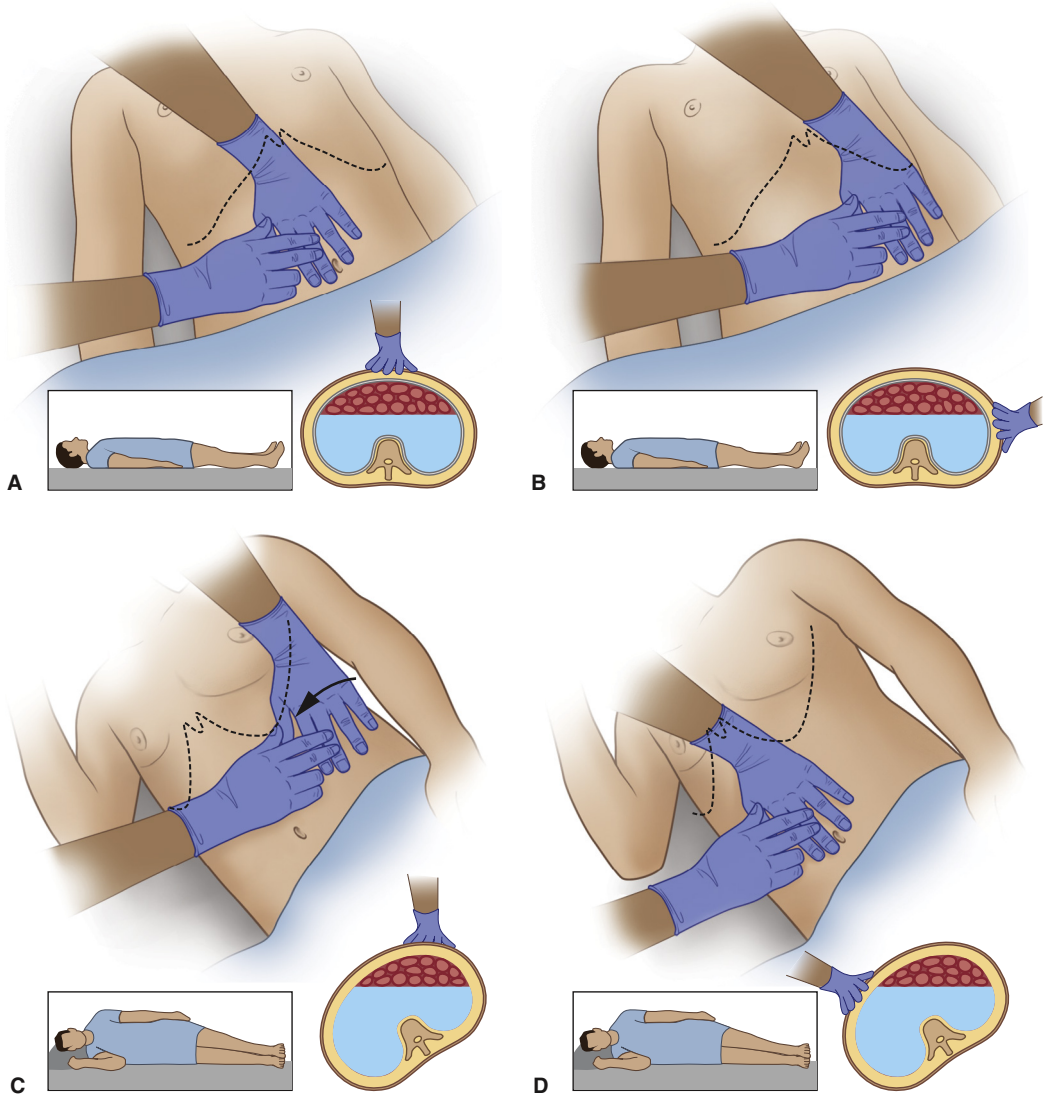


Figure 14.25 Steps in eliciting shifting dullness to confirm presence of ascites on abdominal examination. **(A)** Patient in supine position: Percuss in midline to confirm percussion note is resonant due to gas-filled bowel loops in midline under the hand. Move the percussing hand in four steps to left flank. **(B)** Patient in supine position: Percussion note will change to a dull note due to presence of fluid under hand in left flank. Keep your hand fixed in left flank where you find the percussion note is dull and without moving hand position request patient to turn to right lateral position as in Figure C. **(C)** Patient in right lateral position (see Box): Maintain hand in left flank and percuss after 1 minute: percussion note will change from dull to resonant as fluid shifts from flank to midline and air-filled bowel move under the hand. **(D)** Patient in right lateral position: Move hand in four steps to midline: Percussion note will again change from resonant to dull as fluid is now in midline confirming dullness has shifted from left flank to midline.

If obstruction progresses leading to bowel necrosis, peristalsis ceases and sounds lessen in volume and frequency. In generalized peritonitis, bowel activity rapidly disappears and a state of paralytic ileus ensues, with gradually increasing abdominal distension. The abdomen is 'silent', but one must listen for several minutes before being certain that there are no sounds. Frequently, towards the end of this period, a short run of faint, very high-pitched tinkling sounds is heard. This represents fluid spilling over from one distended loop to another and is characteristic of ileus.

A succussion splash may be heard without a stethoscope and also on auscultation, when there is pyloric stenosis, in advanced intestinal obstruction with grossly distended loops of bowel and in paralytic ileus. This sign is particularly useful in paediatrics. It is generally less used in adults, but should be attempted in patients with chronic vomiting and weight loss, and not just after a meal. It is easiest to elicit without a stethoscope. After explanation, clasp the supine patient's lower ribs and, with your ear near his upper abdomen, give a sharp shake from side to side. A positive sign is when a splashing sound can be heard.



Figure 14.26 Eliciting a fluid thrill. (The hand in the middle of the abdomen is that of an assistant.)

Vascular bruits

Listen for bruits by light application of the stethoscope above and to the left of the umbilicus (aorta), the iliac fossae (iliac arteries), epigastrium (coeliac or superior mesenteric arteries), laterally in the mid-abdomen (renal arteries) or over the liver (increased blood flow in liver tumours—classically primary liver cancer). If an arterial bruit is heard, it is a significant finding that indicates turbulent flow in the underlying vessel, owing to stenosis, aneurysm or a malignant circulation.

Examination of the groin

The groin examination is carried out after examining the abdomen. Before beginning, explain what the examination involves, confirm patient consent, obtain a chaperone, wear gloves and make sure there is adequate light. Proceed to uncover and inspect both groins and the external genitalia. In male patients, inspect and examine the penis and scrotum for any swellings and to confirm that both testes are in their normal position. In the female patient, inspect and examine the external labial folds and the inguinal region for any obvious lumps, observing for general symmetry in structures.

Once the groins have been inspected, ask the patient to turn his head away from you to one side and cough. Look at both inguinal canals for any expansile impulse. If none is apparent, place the left hand in the left groin so that the fingers lie over and in line with the inguinal canal; place the right hand similarly in the right groin (Fig. 14.27). Now ask the patient to give a loud cough and feel for any expansile impulse with each hand. When a patient coughs, the muscles of the abdominal wall contract violently and this imparts a definite, though not expansile, impulse to the palpating hands which is a source of confusion to the inexperienced. Trying to differentiate this normal contraction from a small, fully reducible inguinal hernia is difficult, and the matter can usually be resolved only when the patient is standing up.



Figure 14.27 Palpating the groins to detect an expansile impulse on coughing.

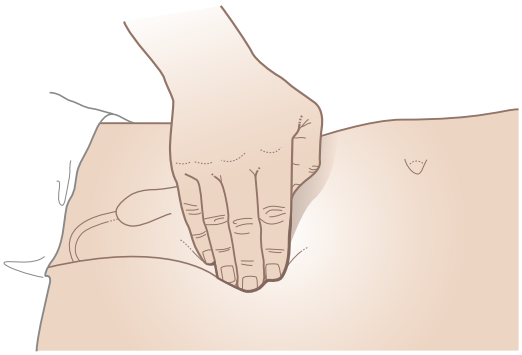


Figure 14.28 Palpation of the femoral vessels.

The common femoral vessels are found just below the inguinal ligament at the mid-point between the anterior superior iliac spine and symphysis pubis. Place the pads of the right index, middle and ring fingers over this site in the right groin and palpate the wall of the vessel. Note the strength and character of its pulsation and then compare it with the opposite femoral pulse (Fig. 14.28). Now palpate along the femoral artery for enlarged inguinal nodes, feeling with the fingers of the right hand, and carry this palpation medially beneath the inguinal ligament towards the perineum. Then repeat this on the left side.

What to do if a patient complains of a lump in the groin

A patient who complains of a lump in the groin should be examined both lying down and standing up. A lump in the groin or scrotum is a common clinical problem in all age groups. Most lumps in the groin are caused either by herniae or enlarged inguinal nodes; inguinal herniae are considerably more common than femoral, with an incidence ratio of 4:1. In the scrotum, hydrocele of the tunica vaginalis or a cyst of the epididymis are common causes of painless swelling; acute epididymo-orchitis is the most frequent cause of a painful swelling. Generalized diseases, such as lymphoma, may present as a lump in the groin. Usually the diagnosis of a lump in the groin or scrotum can be made simply and accurately.

Ask the patient to stand in front of you, get him to point to the side and site of the swelling and note whether it extends into the scrotum. Ask him to turn his head away from you and give a loud cough; look for an expansile impulse and try to decide whether it is above or below the crease of the inguinal ligament. If an expansile impulse is present on inspection, it is likely to be a hernia, so move to whichever side of the patient the lump in the groin is on. Stand beside and slightly behind the patient. If the right groin is being examined, place the left hand over the right buttock to support the patient, the fingers of the right hand being placed obliquely over the inguinal canal. Now ask the patient to cough again. If an expansile impulse is felt, then the lump must be a hernia.

Next decide whether the hernia is inguinal or femoral. The best way to do this is to determine the relationship of the sac to the pubic tubercle. To locate this structure, push gently upwards from beneath the neck of the scrotum with the index finger (Fig. 14.29), but do not invaginate the neck of the scrotum because this would be painful. The tubercle will be felt as a small bony prominence 2 cm from the midline on the pubic crest. In thin patients, the tubercle is easily felt, but this is not so in the obese. If the tubercle is difficult to feel, follow up the tendon of adductor longus, which arises just below the tubercle.

If the hernial sac passes medial to and above the index finger placed on the pubic tubercle, then the hernia must be inguinal in site; if it is lateral to and below, then the hernia must be femoral in site.

If it has been decided that the hernia is inguinal, then one needs to know these further points:

- What are the contents of the sac? Bowel tends to gurgle and is soft and compressible, whereas the omentum feels firmer and is of a doughy consistency.
- Is the hernia fully reducible or not? It is best to have the patient lie down to decide this. Ask the patient if he is 'able to push the hernia back in' and, if so, ask him to do so and confirm yourself. (It is more painful if the examiner reduces it.)
- Is the hernia direct or indirect? Again, it is best to have the patient lie down to decide this. Inspection of the direction of the impulse is often diagnostic, especially in thin patients. A direct hernia tends to bulge straight out through the posterior wall of the inguinal canal, whereas in an indirect hernia the impulse can often be seen to travel obliquely down the inguinal canal. Another helpful point is to place one finger just above the mid-inguinal point over the deep inguinal ring (Fig. 14.30). If a hernia is fully controlled by this finger, it must be an indirect inguinal hernia.

Apart from a femoral hernia, the differential diagnosis of an inguinal hernia includes a large hydrocele of the tunica vaginalis, a large cyst of the epididymis (one should be able to 'get above' and feel the upper border of both of these in the scrotum), an undescended or ectopic testis (there will be an empty scrotum on the affected side), a lipoma of the cord and a hydrocele of the cord.

In considering the differential diagnosis of a femoral hernia, one must think not only of an inguinal hernia but of a lipoma in the femoral triangle, an aneurysm of the femoral artery (expansile pulsation will be present), a saphenovarix (the swelling disappears on lying down, has a bluish tinge to it, varicose veins are often present and there may be a venous hum), a psoas abscess (the mass is fluctuant and may be compressible beneath the inguinal ligament to appear above it in the iliac fossa) and an enlarged inguinal lymph node. Whenever the latter is found, the spine, the feet, legs, thighs, scrotum, perineum and the pudendal and perianal areas must be carefully scrutinized for a source of infection or primary tumour.

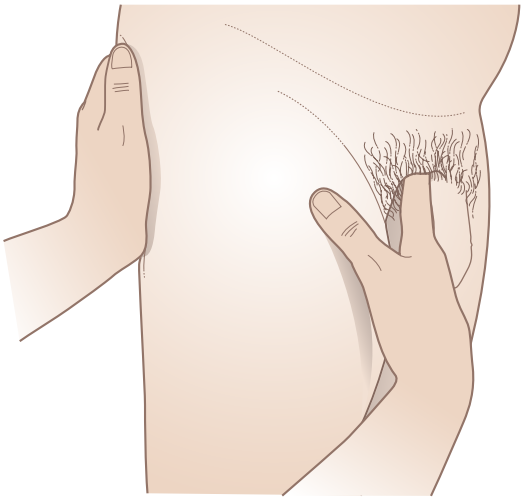


Figure 14.29 Locating the pubic tubercle. Note the position of the examiner, at the side of the patient, with one hand supporting the buttock.

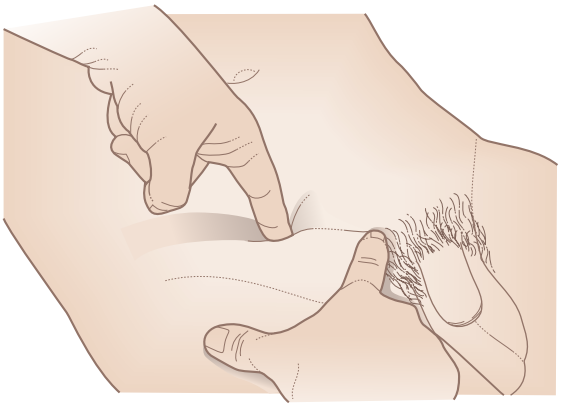


Figure 14.30 Left hand: index finger occluding the deep inguinal ring. Right hand: index finger on the pubic tubercle.

The examination is completed by examining the opposite groin in a similar manner.

External genitalia

Examination of the genitalia is important in patients presenting with abnormalities in the groin and in patients with abdominal symptoms because disease of the genitalia may lead to abdominal symptoms, such as pain or swelling. This is an intimate examination and most patients will find this embarrassing and potentially distressing; it is vital to reassure the patient by ongoing explanation to the patient of what is involved throughout this part of the examination and to stop immediately in case you sense distress. A detailed description of the examination of the male genitalia is given in Chapter 17 and of the female genitalia in Chapter 5.

The anus and rectum

Although examination of the anus and rectum has been widely considered as important component of the examination of the gastrointestinal system, it is essential in patients with suspicion of incontinence, constipation, GI bleed or with peri-anal symptoms. As this is an intimate examination, it requires clear explanation of the steps of the examination in easy to understand language and clear patient agreement and consent, as well as clarifying if the patient is in pain, the need for a chaperone and if the patient is comfortable to proceed.

The left lateral position is best for routine examination of the rectum (Fig. 14.31). Make sure that the buttocks project over the side of the couch with the knees drawn well up and that a good light is available. Once patient consent is obtained, don disposable gloves and clinical apron, as well as having lubrication jelly and tissues close at hand. Stand behind the patient's back, facing the patient's feet. Reiterate briefly to the patient what you are about to do, that you will be as gentle as possible and that you will stop the examination, if requested, at any time.

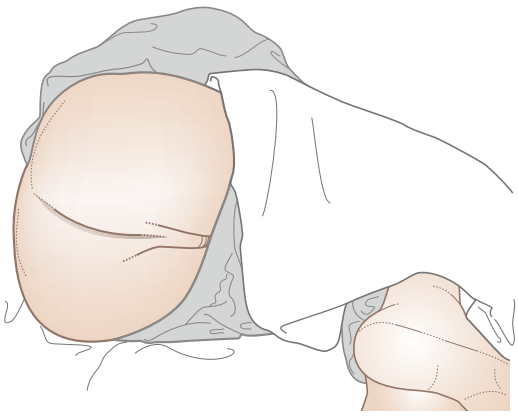


Figure 14.31 Left lateral position for rectal examination.

Inspection

Separate the buttocks carefully and inspect the perianal area and anus. Note the presence of any abnormality of the perianal skin, such as inflammation or excoriation, which may vary in appearance from mild erythema to a raw, red, moist, weeping dermatitis or, in chronic cases, thickened white skin with exaggeration of the anal skin folds. The latter form anal skin tags, which may follow not only severe pruritus but also occur when prolapsing haemorrhoids have been present over a period of time. Tags should not be confused with anal warts (condylomata acuminata), which are sessile or pedunculated papillomata with a red base and a white surface. Anal warts may be so numerous as to surround the anal verge and even extend into the anal canal. Note any 'hole' or dimple near the anus with a telltale bead of pus or granulation tissue surrounding it, which represents the external opening of a fistula-in-ano. Usually it is easy to distinguish a fistula-in-ano from a pilonidal sinus, in which the opening lies in the midline of the natal cleft but well posterior to the anus.

A number of painful anorectal conditions usually can be diagnosed readily on inspection. An anal fissure usually lies directly posterior in the midline. The outward pathognomonic sign of a chronic fissure is a tag of skin at the base (sentinel haemorrhoid). If pain allows, the fissure can easily be demonstrated by gently drawing apart the anus to reveal the tear in the lining of the anal canal.

A perianal haematoma (thrombosed external haemorrhoid) occurs as a result of rupture of a vein of the external haemorrhoidal plexus. It is seen as a small (1 cm), tense, bluish swelling on one aspect of the anal margin and is exquisitely tender to the touch. In prolapsed strangulated haemorrhoids, there is gross swelling of the anal and perianal skin, which looks like oedematous lips, with a deep red or purple strangulated haemorrhoid appearing in between, and sometimes partly concealed by, the oedema of the swollen anus. In a perianal abscess, an acutely tender, red, fluctuant swelling is visible which deforms the outline of the anus. Usually it is easy to distinguish this from an ischiorectal abscess where the anal verge is not deformed, the signs of acute inflammation are often lacking and the point of maximum tenderness is located midway between the anus and ischial tuberosity.

Note the presence of any ulceration. Finally, if rectal prolapse is suspected, ask the patient to bear down (as if trying to pass stool) and note whether any pink rectal mucosa or bowel appears through the anus or whether the perineum itself bulges downwards. Downward bulging of the perineum during straining on bending down or in response to a sudden cough indicates weakness of the pelvic floor musculature, usually owing to denervation of these muscles. This sign is often found in women after childbirth, in women with faecal or urinary incontinence and in patients with severe chronic constipation.

Digital rectal examination (palpation)

Put a generous amount of lubricant (jelly) on the gloved index finger of the right hand, place the pad of the finger (not the tip) flat on the anus (Figs 14.32 and 14.33) and press firmly and slowly (flexing the finger) in a slightly backwards direction. After initial resistance, the anal sphincter relaxes and the finger can be passed into the anal canal. If severe pain is elicited when attempting this manoeuvre, then further examination should be abandoned because it is likely the patient has a fissure and the rest of the examination will be very painful and unhelpful.

Feel for any thickening or irregularity of the canal wall, making sure that the finger is carefully turned through a full circle (180° each way). Assess the tone of the anal musculature; it should normally grip the finger firmly. If there is any doubt, ask the patient to contract the anus on the examining finger by asking to 'squeeze as if trying to stop passage of urine or faeces'. A cough will induce a brisk contraction of the external anal sphincter which should be readily appreciated. In the old and infirm with anal incontinence or prolapse, almost no appreciable contraction will be felt. With experience it is usually possible to feel a shallow groove just inside the anal canal which marks the dividing line between the external and internal sphincter. The anorectal ring may be felt as a stout band of muscle surrounding the junction between the anal canal and rectum.

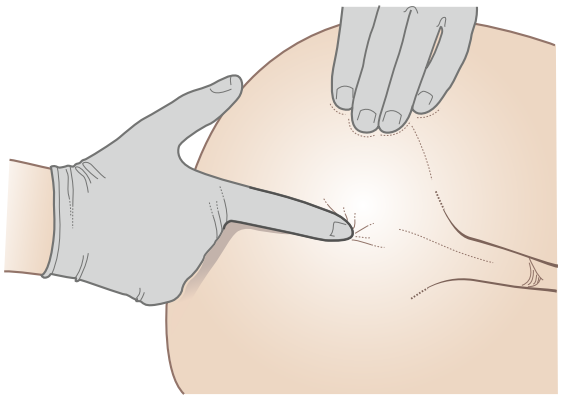


Figure 14.32 Correct method for insertion of the index finger in rectal examination. The pad of the finger is placed flat against the anus.

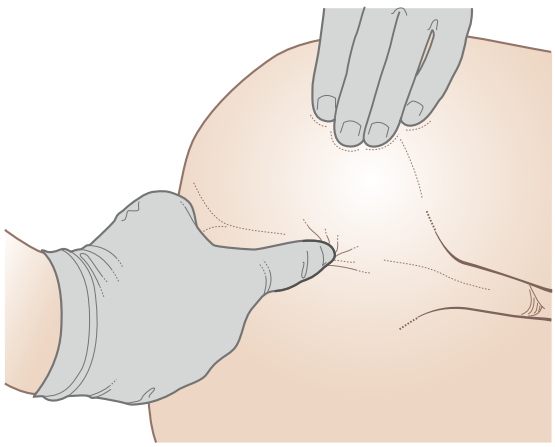


Figure 14.33 Incorrect method of introduction of the finger into the anal canal.

Now pass the finger into the rectum. The examiner's left hand should be placed on the patient's right hip and later it can be placed in the suprapubic position to exert downward pressure on the sigmoid colon. Try to visualize the anatomy of the rectum, particularly in relation to its anterior wall. The rectal wall should be assessed with sweeping movements of the finger through 360°, 2, 5 and 8 cm inwards or until the finger cannot be pushed any higher into the rectum. Repeat these movements as the finger is being withdrawn. In this way it is possible to detect malignant ulcers, proliferative and stenosing carcinomas, polyps and villous adenomas. The hollow of the sacrum and coccyx can be felt posteriorly. Laterally, on either side, it is usually possible to reach the side walls of the pelvis. In men, one should feel anteriorly for the rectovesical pouch, seminal vesicles (normally not palpable) and the prostate. In a patient with a pelvic abscess, however, pus gravitates to this pouch, which is then palpable as a boggy, tender swelling lying above the prostate. Malignant deposits will feel hard and, in infection of the seminal vesicles, these structures become palpable as firm, almost tubular swellings deviating slightly from the midline just above the level of the prostate.

Assessment of the prostate gland is important. It forms a rubbery, firm swelling about the size of a large walnut. Run the finger over each lateral lobe, which should be smooth and regular. Between the two lobes lies the median sulcus, which is palpable as a faint depression running vertically between each lateral lobe. Although it is possible to say on rectal examination that a prostate is enlarged, accurate assessment of its true size only comes with a lot of experience. In carcinoma of the prostate, the gland loses its rubbery consistency and becomes hard, whereas the lateral lobes tend to be irregular and nodular with distortion or loss of the median sulcus.

The cervix is felt as a firm, rounded mass projecting back into the anterior wall of the rectum. This is often a disconcerting finding for the inexperienced. The body of a retroverted uterus, fibroid mass, ovarian cyst, malignant nodule or a pelvic abscess may all be palpated in the pouch of Douglas (rectouterine pouch), which lies above the cervix. This aspect of the rectal examination forms an essential part of pelvic assessment in female patients.

On withdrawing the finger after rectal examination, look at it for evidence of mucus, pus and blood, either fresh or altered. If in doubt, wipe the finger on a white swab. Finally, make sure to wipe the patient

clean before telling the patient that the examination is completed and also tell him to be careful as he rolls to the supine position as he will be very near the edge of the couch or bed.

Nutritional assessment

Nutritional assessment is important in all patients because most chronic and acute diseases affect nutrition and nutritional status has a key influence on prognosis of patients.

The following aspects constitute nutritional assessment:

1. Documenting a history of weight loss: The simplest nutritional assessment is to ask about weight loss and what the patient's weight was before the illness in all patients, including those who are obese. Patients at risk of weight loss and malnutrition may have GI disease preventing eating, reducing appetite or preventing absorption of nutrients or non-GI disease causing reduced appetite (especially malignancy). Increased energy consumption is also important in some patients with cancer and those with severe sepsis, thyrotoxicosis or burns.
2. Dietary history: Full medical clerking in a patient who has lost weight needs to include a simple assessment of the quantity and variety of foods eaten, as well as any restrictions on eating (e.g. poor dentition, social and financial circumstance) or special diets followed for medical reasons (e.g. a gluten-free diet in coeliac disease). If there are concerns then referral to a dietician for detailed diet history and a diet diary will be required.
3. Full examination including looking for signs of general and nutrient-specific malnutrition, see details in Table 14.6. Body weight and patient height are key parts of the general examination.

Table 14.6 Principal symptoms and signs caused by vitamin and mineral deficiencies

Nutrients	Deficiency syndrome	Principal symptoms/signs
Vitamin A, retinol (carotenoids)	Protein-energy malnutrition	Night blindness, Bitot spots, keratomalacia
Vitamin B_1, thiamine	Wernicke/Korsakoff, beri-beri	Nystagmus, sixth cranial nerve palsy, ataxia (Wernicke's encephalopathy) Symmetrical peripheral neuropathy (dry beri-beri), fulminant cardiac failure, lactic acidosis (Shoshin beri-beri), high output cardiac failure (wet beri-beri)
Vitamin B_2, riboflavin	Ariboflavinosis	Angular stomatitis, glossitis, magenta tongue
Vitamin B_3, niacin, nicotinic acid	Pellagra	Dermatitis of sun-exposed areas (including 'Casal's Necklace'), dementia, poor appetite, difficulty sleeping, diarrhoea
Vitamin B_6, pyridoxine	Deficiency associated with isoniazid use	Poor appetite, lassitude, oxaluria, seborrhea, neuropathy
Pantothenic acid		Nausea, abdominal pain, paraesthesiae, burning feet
Biotin		Dermatitis, depression, lassitude, muscle pains, electrocardiogram abnormalities, blepharitis
Folic acid	Deficiency associated with methotrexate use	Macrocytic anaemia, thrombocytopenia and megaloblastic bone marrow
Vitamin B_{12}	Pernicious anaemia	Subacute combined degeneration of the spine, macrocytic anaemia
Vitamin C, ascorbic acid	Scurvy	Poor wound healing, fatigue, limb pain, shortness of breath, difficulty sleeping, gingivitis, perifollicular purpura, hyperkeratosis
Vitamin D, ergo-/cholecalciferol	Rickets/osteomalacia	Bone pain, proximal myopathy, waddling gait, growth retardation
Vitamin E, tocopherol		Haemolysis, posterior column signs, ataxia, muscle wasting, retinitis pigmentosa-like changes, night blindness
Vitamin K, phylloquinone and other menaquinones	Chronic liver disease	Bruising, purpura, coagulopathy
Trace elements		
Iron		Koilonychia, smooth tongue, anaemia, ooesophageal web, impaired cognitive development
Zinc	Acrodermatitis enteropathica	Peristomal/perinasal/perineal erythema, thin hair, diarrhoea, apathy, anorexia, growth failure, hypoglycaemia
Copper		Microcytic hypochromic anaemia, neutropenia, scurvy-like bone lesions, osteoporosis
Chromium		Peripheral neuropathy, hyperglycaemia
Selenium		Cardiomyopathy
Iodine		Goitre

Table 14.7 World Health Organization classification of body weight

Category	BMI/Quetelet index
Underweight	<18.5
Healthy weight	18.5–24.9
Overweight	25–29.9
Moderately obese	30–34.9
Severely obese	35–39.9
Morbidly obese	>40

The two values are used to calculate the body mass index (BMI or Quetelet index). This is defined by body weight (in kilograms), divided by height (in metres) squared. The World Health Organization (WHO) classification of this index is given in Table 14.7. In the UK, the range 20–25 is often regarded as desirable, but the lower level of 18.5 is more applicable internationally. Patients with a BMI > 30 are at risk of a higher burden of health conditions and, therefore, are advised to undertake a structured weight loss program with full multidisciplinary team support in order to reduce their long-term health risks. These BMI values have been revised down for patients of South East Asian origin to reflect their higher risk of developing metabolic syndrome and cardiovascular complications at lower BMI. The state of nourishment can be assessed further by the more specialized measurements of mid-upper arm circumference (MUAC), skin-fold thickness, waist circumference and waist-to-hip ratio.

4. Malnutrition Universal Screening Tool (MUST): All UK hospital inpatients are now assessed for their risk of malnutrition. The most common assessment method used by health care staff is the MUST. This involves calculating the patient's BMI and then adding on a score for the percentage of unexpected weight loss and a score for the degree of acute illness. The final score is used to trigger a full dietary assessment and initiation of treatment in those who need it. Patients should be screened weekly whilst in hospital and at every outpatient attendance. It is important to be aware of the limitations of this tool, particularly in patients in whom there has been a lot of fluid retention which may have masked underlying weight loss.

Investigations: (see Box 14.10)

A large variety of investigations can be relevant to patients with GI diseases, including laboratory assessment of bloods, stool and ascitic fluid, endoscopic investigations and imaging as below (see Box 14.10). The choice and sequence of investigation is determined by the symptoms, suspected diagnosis and clinical presentation. A detailed description of each investigation is beyond the scope of this chapter, but it is important to be aware of the range available as a non-specialist as patients may refer to these while narrating their history.

Laboratory-based investigations (see Box 14.10)

Blood tests

Most patients seen in primary care and in hospital settings including within emergency departments have a 'routine' panel of blood tests that include full blood count, liver function tests, urea and electrolytes. Other blood tests may be added based on the diagnosis suspected as follows.

Blood tests for patients with liver disease

Liver function tests (LFTs) are usually included as part of a 'routine' panel of tests in most healthcare settings. If these are abnormal, a further panel of tests is usually required, often referred to as a liver screen. LFTs include serum bilirubin, alkaline phosphatase, serum alanine aminotransferase (ALT), serum aspartate aminotransferase (ALT), gamma glutamyl transferase (GGT) and serum albumin. In addition, prothrombin time is requested in any patient with abnormal LFTs, because it is a sensitive marker of the severity of acute and chronic liver injury.

Patients with abnormal LFTs and sometimes those with normal LFTs but with liver-related risk factors undergo a liver screen. This is a panel of tests that assesses the cause of liver disease and are requested in any patient suspected to have a liver problem, including those who may be asymptomatic and are found to have incidentally detected abnormal LFTs. The commonly used liver screen is shown in Box 14.11 as described below:

Viral hepatitis B and C are associated with chronic liver disease, whereas hepatitis A, B and E are associated with acute liver disease. Patients with chronic liver disease are screened for hepatitis B by testing for hepatitis B surface antigen (HBsAg); hepatitis C, D, E are screened for with antibody testing. If these tests are positive, then confirmatory tests include polymerase chain reaction (PCR) test for hepatitis B DNA, and hepatitis C, D and E RNA, respectively. Patients with acute liver injury are assessed by screening for IgM antibody (for hepatitis A, B, D and E), and confirmation by PCR testing. Hepatitis B assessment involves in addition detection of hepatitis e antigen and e antibody, which provide information about ongoing disease activity.

Box 14.10 Investigations in GI diseases including liver diseases

Laboratory-based investigations

1. Blood tests
2. Stool examination
   - Stool inspection
   - Stool microbiology, culture and sensitivities (MC&S)
   - Faecal occult blood
   - Faecal immunochemical testing (FIT)
   - Faecal calprotectin
   - Faecal elastase
   - Stool *H. pylori*
3. Ascitic fluid examination
   - Biochemistry: protein, albumin, amylase and SAAG (serum albumin ascites gradient)
   - Cell count
   - Malignant cytology
   - Culture

Endoscopic investigations

1. Upper GI endoscopy
2. Enteroscopy and wireless capsule endoscopy
3. Colonoscopy and sigmoidoscopy
4. Endoscopic retrograde cholangiopancreatography (ERCP)
5. Endoscopic ultrasound
6. Percutaneous transhepatic cholangiography

Imaging

1. Plain radiograph
2. Abdominal ultrasound
3. Transient liver elastography
4. Computed tomography (CT)
5. Positron emission tomography scanning (PET)
6. Magnetic resonance imaging (MRI and MRCP)
7. Barium studies
   - Barium swallow
   - Barium follow-through study
   - Small bowel enema
   - Barium enema

Other diagnostic and therapeutic investigations

1. Investigation for liver diseases
   - Liver biopsy
   - Hepatic venography, transjugular biopsy and transjugular intrahepatic porto-systemic shunt
2. Investigation for pancreatic diseases
   - Serum lipase
   - Faecal elastase
   - Pancreatic function test
3. Investigation for oesophageal diseases
   - Manometry
   - pH monitoring
4. Investigation for *H. pylori*
   - *H. pylori* antigen in stool
   - Endoscopic: Rapid urease test, culture
   - ^{14} Carbon labelled breath test
5. Investigation for complicated GI ulcer disease
   - Serum gastrin
6. Investigation for ongoing GI bleed
   - CT angiography of gastrointestinal arteries
   - Technetium tagged red cell scanning
7. Investigation to localize inflammation
   - Radioisotope scanning

Box 14.11 Liver screen: to determine cause of liver disease

Viral liver diseases

- *HAV:* IgM antibody to HAV
- *HBV:* Hepatitis B surface antigen (screening), HBV DNA, HBe antigen and HBe antibody for staging, isolated HB core antibody: marker of previous HBV exposure
- *HCV:* Hepatitis C antibody (screening), HCV RNA (diagnostic)
- *HDV:* Only in presence of HBV infection
- *HEV*: Hepatitis E antibody (screening), HEV RNA (diagnostic)

Immune liver diseases

- *Autoimmune hepatitis:* Elevated IgG and positive anti-nuclear antibody/anti-smooth muscle/anti-liver kidney microsomal antibody
- *Primary biliary cholangitis:* Elevated IgM and positive anti-mitochondrial antibody positive M2 subtype (diagnostic)
- *Primary sclerosing cholangitis:* Anti-neutrophil cytoplasmic antibody positive (ANCA)

Metabolic liver diseases

- *Alcohol-related liver disease*: History of alcohol excess, elevated mean corpuscular volume (MCV)
- *Non-alcoholic fatty liver disease*: Screen for presence of metabolic syndrome: T2DM, dyslipidaemia, central obesity and hypertension
- *Hereditary haemochromatosis*: Elevated serum ferritin and transferrin saturation (screening), HFE gene testing (diagnostic, so called because of H for high and Fe is the symbol for iron)
- *Wilson's disease*: Low serum caeruloplasmin (screening), elevated 24-hr urinary copper excretion (diagnostic)
- *Alpha 1 antitrypsin deficiency (A1AT):* Low levels (screening)

Tests to screen for autoimmune liver disease include antibodies to antinuclear antigen (ANA), smooth muscles (SMA) and liver kidney microsomes, which if present suggest autoimmune hepatitis. Anti-mitochondrial antibodies are found in the blood in primary biliary cholangitis in over 90% of cases.

A major cause for liver disease is alcohol. The most reliable screen for this remains an accurate history, documenting weekly consumption of alcohol in units (regular consumption of alcohol greater than 14 units per week is a risk for alcohol-related liver disease). Elevated mean corpuscular volume (MCV), elevated GGT and an ALT-to-AST ratio less than 1 are more often seen in patients with alcohol-related liver disease. It is also important to screen for the presence of alcohol dependence in patients with suspected alcohol-related liver disease using various validated tools (see Box 14.12).

Box 14.12 Simple screening tools for alcohol

CAGE: The CAGE uses four simple questions to assess alcohol use. Importantly, the questions should not be preceded by any questions about alcohol use; its sensitivity is increased by an open-ended introduction:

1. Have you ever felt the need to *C*ut down on your drinking?
2. Have people *A*nnoyed you by criticising your drinking?
3. Have you ever felt bad or *G*uilty about your drinking?
4. Have you ever had a drink first thing in the morning to steady your nerves or get over a hangover (*E*ye opener)?

A score of 2 or more is considered clinically significant

AUDIT: The alcohol use disorders identification (AUDIT) test is a longer screening tool recommended by the World Health Organisation.

Screen for other liver diseases include screening for liver disease owing to excess iron (hereditary haemochromatosis) or copper accumulation (Wilson's disease) and can be detected by measurements of elevated serum ferritin or low levels of serum caeruloplasmin, respectively. In primary haemochromatosis, genetic studies are used to confirm the diagnosis. Confirmation of Wilson's disease requires measurement of 24-hour urinary copper excretion and careful examination for copper deposition in the iris (Kayser-Fleischer ring) by slit-lamp examination.

Another common cause for chronic liver disease is non-alcoholic fatty liver disease (NAFLD), which is associated with the metabolic syndrome and screening for hypertension, dyslipidemia, central obesity and type 2 diabetes, and excluding other causes of liver disease, is a useful tool for diagnosing NAFLD.

Anaemia

Anaemia is a common finding in patients with GI and liver disease and an initial assessment will include measuring serum ferritin, vitamin B_{12} and folate levels and undertaking a coeliac screen (see below).

Coeliac screen

These include antibodies to tissue transglutaminase, endomysial and gliadin antigen. As these are IgA antibodies, it is important that IgA levels are checked at the same time to exclude IgA deficiency, which could lead to a false-negative result.

Weight loss screen

In addition to blood tests, as above, this includes thyroid function tests and erythrocyte sedimentation rate (ESR).

Patients with inflammatory bowel disease

In addition to routine blood tests, C-reactive protein (CRP) is used to assess the severity of inflammation and response to therapy.

Stool inspection used to be part of bedside assessment and is carried out less often owing to concerns regarding specimen handling and infection control issues. It may be possible to get a description of stools from the history. Watery stools are found in all cases of profuse diarrhoea and after administration of purgatives as well as antibiotic-associated diarrhoea in which case *Clostridium difficile* infection needs to be excluded. In cholera, the stools—known as rice-water stools—are colourless, almost devoid of odour, alkaline in reaction and contain a number of small flocculi consisting of shreds of epithelium and particles of mucus. Purulent or pus-containing stools are found in severe infectious dysentery or ulcerative colitis. Slimy stools are owing to the presence of an excess of mucus and point to a disorder of the large bowel. The mucus may envelop the faeces or may be intimately mixed with it. Presence of blood in stools varies in appearance according to the site of bleed within the GI tract and the volume of blood loss over time. If the bleeding takes place in the upper GI tract, the stools are black, shiny and foul smelling, called melaena. Red-currant jelly stools are classically described in children presenting with small bowel intussusception. Large bowel bleeding proximal to the rectum may produce darker red blood in stools, whereas bleeding within the rectum or anus is often brighter red and referred to as exit type per rectal (PR) bleed. There is blood streaking the side of faeces anal fissure and haemorrhoidal bleeding may be found mainly on the toilet paper or as drops of bright red blood in the pan. A brisk upper-GI bleed may lead to bright red rectal bleeding (haematochezia). The stools of bacillary dysentery initially consist of faecal material mixed with blood and pus, later of blood and pus without faecal material. Those of amoebic dysentery characteristically consist of fluid faecal material, mucus and small amounts of blood. The stools of steatorrhoea are very large, pale and putty- or porridge-like, sometimes frothy with a visible oily film, and often float. They are apt to stick to the sides of the toilet and are difficult to flush away.

Reliable microbiology, culture and sensitivity (MC&S) of the stool is vital for accurate diagnosis of acute and chronic diarrhoea. All stool samples for microbiological examination should be sent to the laboratory as fresh as possible; if amoebic or similar infection is suspected, then microscopy should be performed with the stool still as close to body temperature as possible (a 'hot' stool). Any microbiological finding must be correlated with the history, because many patients are healthy carriers of organisms that can be pathogenic in other situations. If *C. difficile* infection is suspected in a patient with diarrhoea following the use of antibiotics, the laboratory must be asked to look for the *Clostridium* toxin in preference to the organism.

The guaiac test (haemoccult) previously has been the most widely used method for testing for faecal occult blood. A filter paper impregnated with guaiac

turns blue in the presence of haemoglobin when hydrogen peroxide is added. The test depends on the oxidation of guaiac in the presence of haemoglobin. Other substances with peroxidase activity, including dietary substances such as bananas, pineapple, broccoli and radishes, can produce a false-positive reaction, and ascorbic acid may cause false-negative results. Therefore, dietary preparation is necessary for accurate screening, for example, when screening a population in the early detection of colonic cancer. In ordinary clinical use, the test is sensitive to faecal blood losses of about 20 ml per day. This test can be used for patients on a normal diet, but it may not detect small amounts of GI bleeding. The test may be negative in the presence of lesions that bleed intermittently or slightly, particularly those situated in the upper GI tract.

Faecal immunofaecal (FIT) testing is increasingly used to check for the presence of occult blood in stool sample instead of above and is being rapidly adopted for use in primary care to screen patients with bowel symptoms or iron deficiency anaemia who may require referral to secondary care for further investigations including endoscopic assessment.

The faecal calprotectin test is used to assess for the presence of bowel inflammation and is most often used in primary care to differentiate inflammatory bowel disease from irritable bowel syndrome in younger patients. If elevated, it should prompt referral to secondary care for further investigations, including colonoscopy. It may also be elevated in patients taking nonsteroidal anti-inflammatory drugs.

Faecal elastase has largely replaced measurement of faecal fat in patients suspected to have diarrhoea owing to malabsorption from pancreatic diseases (steatorrhoea) and, if reduced, may indicate pancreatic exocrine insufficiency (although measurements can vary over time).

Helicobacter pylori is a common upper GI pathogen and it is associated with chronic gastritis, duodenitis and peptic ulcer disease. This stool test is used mainly in primary care to check for the presence of the infection, and often leads to treatment without needing endoscopy.

Ascitic fluid examination

Ascitic fluid sampling is carried out at the bedside taking a blood sample and is analysed for biochemistry, cell count, cytology and culture. It is indicated in all patients with new clinically apparent ascites to help ascertain its cause. All patients known to have ascites from liver cirrhosis who are admitted to hospital require an ascitic fluid examination to rule out the presence of spontaneous bacterial peritonitis which is potentially life-threatening. Ascitic fluid is sampled through the abdominal wall similarly to taking a blood sample.

Biochemical analysis includes measurement of total protein, albumin, amylase and a calcluation of the serum ascites albumin gradient (SAAG). Ascitic transudates, such as those that occur in heart failure, cirrhosis and nephrotic syndrome, normally have a protein content under 25 g/l (i.e. less than two-thirds the concentration of albumin in the plasma). Exudates occurring in tuberculous peritonitis or in the presence of secondary malignancy usually contain more than 25 g/l of protein. This method of distinction, however, is somewhat unreliable as it does not take into account the associated serum albumin levels. Therefore, it is now routine to measure serum albumin and ascitic fluid albumin to calculate the SAAG; a value of ≥11 g/l suggests that the ascites is secondary to portal hypertension (liver cirrhosis). If pancreatic ascites is suspected the amylase content of the fluid should be measured.

Lymphocytes in ascitic fluid are characteristic of chronic inflammation, the most common cause being tuberculous peritonitis, but acid-fast bacilli are often not seen on staining. In contrast, an elevated neutrophil count is seen in the presence of acute peritonitis. Very high neutrophil counts (>10,000 per mm^3) in ascitic fluid are classically seen in patients with peritonitis secondary to perforation of the bowel. This needs to be differentiated from spontaneous bacterial peritonitis, a condition often seen in patients with liver cirrhosis who are suddenly less well. The neutrophil count is lower, but organisms may not be cultured: the diagnosis is primarily based on the finding of >250 or more neutrophils per mm^3 fluid, results which should lead to the use of prompt and appropriate antibiotics.

In all new cases of ascites, the fluid should undergo careful examination for the presence of malignant cells, particularly if an underlying abdominal malignancy is suspected and/or ascitic aspiration reveals blood-stained fluid. (Fig. 14.34).

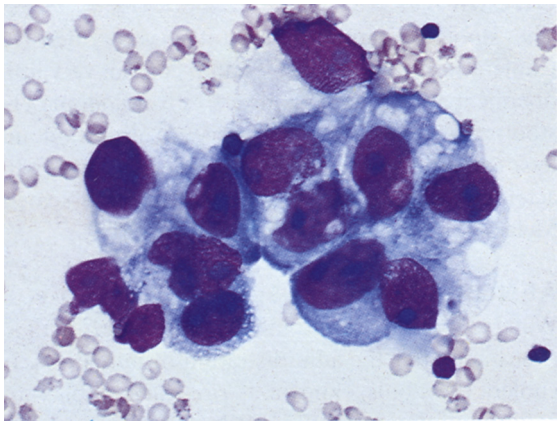


Figure 14.34 Ascites cytology. A group of tumour cells showing random orientation and large abnormal nucleoli indicating malignancy. Ascitic fluid from patient with ovarian carcinoma. May-Grünwald-Giemsa stain ×160.

Ascitic fluid cultures are positive in patients with peritonitis secondary to bowel perforation. However, in spontaneous bacterial peritonitis, the cultures are often negative and therefore in patients with ascites owing to cirrhosis, inoculating 10 ml of ascitic fluid sample into blood culture bottles may increase the yield.

Endoscopic investigations (see Box 14.10)

1. Upper GI endoscopy
2. Enteroscopy and wireless capsule endoscopy
3. Colonoscopy and sigmoidoscopy
4. ERCP
5. Endoscopic ultrasound
6. Percutaneous transhepatic cholangiography

Upper gastrointestinal endoscopy

In the last 50 years or so, the development of the fibreoptic endoscope and, more recently, of video-endoscopy, has revolutionized the assessment of the upper GI tract (Fig. 14.35). With these instruments it is possible to inspect directly as far as the proximal small bowel, with or without conscious sedation and local pharyngeal anaesthesia. Because of the ability to photograph and biopsy any suspicious lesions, this technique is the investigation of choice for demonstrating structural abnormalities in the upper gut. Therapeutic endoscopy is now the treatment of choice for bleeding varices, oesophageal obstruction and, in most cases, of bleeding peptic ulcer (Fig. 14.36). In inoperable cancer of the oesophagus, palliative prostheses (stents) can be inserted endoscopically. Enteral nutrition may be delivered via percutaneous endoscopic gastrostomy tubes.

Enteroscopy and wireless capsule endoscopy

Endoscopic examination of the small bowel, which requires the use of a longer small bowel endoscope, is called enteroscopy. Samples of small intestine mucosa are valuable for histological diagnosis in various types of malabsorption. Direct biopsy of the lower duodenum at conventional upper GI endoscopy is usually sufficient for diagnosis of coeliac disease, but jejunal tissue can also be obtained from the first few small bowel loops using a push-type flexible enteroscopes. If coeliac disease is suspected, a coeliac blood screen is requested (as described), but a duodenal biopsy is essential to complete or fully exclude the diagnosis. The low-power view of the villi is as important as the detailed histology. Other malabsorption problems, such as Whipple's disease and chronic giardiasis, may also be seen on duodenal biopsy. Very occasionally, full-thickness jejunal biopsies are used for diagnosing neuromuscular gut disorders and can be obtained by laparoscopy. These tests are limited in view of the long length of the small bowel, but more recently virtual capsule endoscopy has evolved. Wireless capsule endoscopy provides excellent endoscopic images of the whole of the small bowel, without the need for intubation, but, as yet, biopsies are not possible by this method.

Colonoscopy and sigmoidoscopy

As in the upper gut, the use of flexible fibreoptic and video instruments has revolutionized the investigation of the colon. Sigmoidoscopy involves examination of the proximal sigmoid, rectum and anus, after a simple enema preparation, whereas colonoscopy allows examination up until the caecum and distal ileum and requires more extensive colon preparation. It is also an outpatient procedure, under conscious sedation. These techniques are invaluable for diagnosing and obtaining tissue for confirming inflammatory and neoplastic disease and for removal of all types of polyps (Figs 14.37 and 14.38). Dilation of strictures,

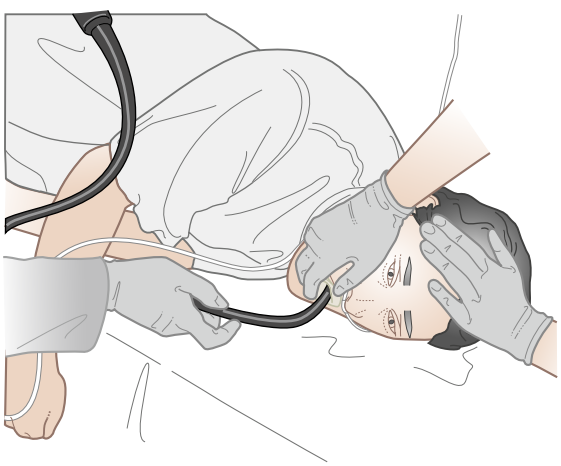


Figure 14.35 A patient undergoing endoscopy of the upper GI tract. The plastic guard between the teeth prevents the instrument from being bitten. The patient may have received pharyngeal local anaesthetic spray or light sedation with midazolam.

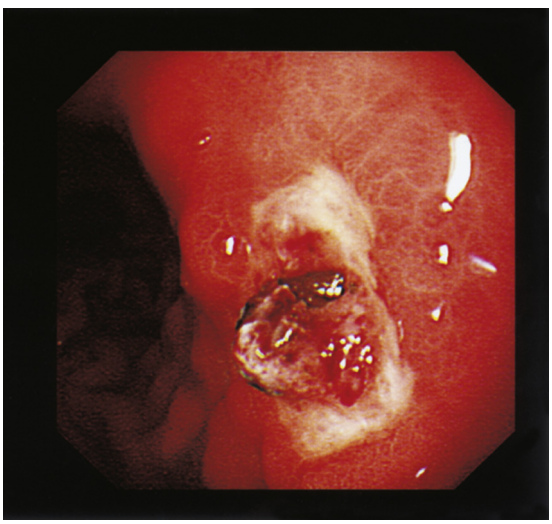


Figure 14.36 Endoscopic view of a duodenal ulcer that has recently bled with a visible blood vessel on its surface.

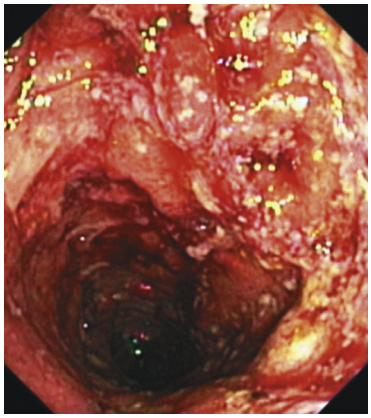


Figure 14.37 View of colonic epithelium at colonoscopy, showing severe ulcerative colitis with extensive ulceration and bleeding.

stenting or other thermal treatments of obstructing tumours can also be applied during endoscopy. A skilled colonoscopist should reach the caecum in more than 90% of examinations attempted and the terminal ileum in 50%. There is a small risk (1:1000) of colonic perforation associated with the colonoscopy. Sigmoidoscopy is particularly useful in the differential diagnosis of diarrhoea of colonic origin. Granularity, loss of vascular pattern and ulceration with bleeding may indicate the presence of ulcerative colitis; aphthous ulceration may suggest Crohn's disease; and multiple rounded white macules may be diagnostic of pseudomembranous colitis caused by C. *difficile* toxin, usually following antibiotic treatment. In suspected amoebic dysentery, the mucous membrane should be inspected and portions of mucus and scrapings from the ulcerated mucosa may be removed and examined microscopically for amoebic cysts.

The anal canal and lower rectum can be readily visualized with a rigid proctoscope. Place the patient in the position described for rectal examination and gently pass the lubricated instrument to its full depth. Remove the obturator and inspect the mucosa as the instrument is slowly withdrawn and air gently insufflated. Haemorrhoids are seen as reddish/blue swellings which bulge into the lumen of the instrument. Asking the patient to strain down as the proctoscope is withdrawn exaggerates any haemorrhoids. The internal opening of an anal fistula, an anal or low rectal polyp and a chronic anal fissure are other abnormalities that may be seen.

Endoscopic retrograde cholangiopancreatography (ERCP)

This technique is useful in the rapid diagnosis and localization of the different causes of jaundice owing to obstruction of the main bile ducts. Using a special side-viewing duodenoscope, the duodenal papilla is identified and a cannula passed through it into the common bile duct. Radiopaque contrast is then injected into the cannula and the whole of the biliary system is visualized by X-ray imaging. Needle or forceps biopsy and brush cytology may give a specific diagnosis of strictures of the biliary tree. ERCP has an important therapeutic role in the treatment of jaundice because it allows the removal of bile duct stones or the placement of stents (tubes that facilitate the passage of bile into the duodenum past obstructing lesions, such as tumours of the pancreas or bile duct). More novel techniques allow direct visualization of the biliary system. Such therapies often involve performing a sphincterotomy during ERCP using a cutting diathermy wire passed into the bile duct via the ampulla. The sphincterotomy opens the ampulla and often allows the removal of stones in the bile duct. Rapid advances have been made in this field with dedicated mother and baby cholangioscopes, which provide greater accuracy and range of options to help in diagnosis, tissue biopsy and biliary interventions.



Figure 14.38 Colonoscopic view of a tubulovillous adenoma.

The entire pancreatic duct system can be visualized by ERCP. Therefore, the technique is valuable in the diagnosis of chronic pancreatitis. In patients with pancreatic carcinoma, a needle biopsy can be performed by ERCP, and brush cytology of the pancreatic duct may also provide histological confirmation of the diagnosis. ERCP carries a small mortality and may be complicated by pancreatitis, bleeding, perforation or infection. Prophylactic antibiotics should be used if there is biliary obstruction, and every procedure should be preceded by a large dose of nonsteroidal anti-inflammatory drugs (usually by suppository) to reduce the risk of pancreatitis afterwards. The advent of magnetic resonance imaging of the bile and pancreatic ducts now means that most ERCPs are therapeutic.

Endoscopic ultrasound

Endoscopic ultrasound is a relatively new tool that uses endoscopic access to carry out ultrasound assessment of luminal organs including the oesophagus, stomach, bowel and biliary tree. Endoscopic ultrasound is more sensitive in staging the mucosal depth of penetration of cancers in the oesophagus and stomach than computed tomography (CT) scanning; it is frequently better at lymph node detection, but is less effective at detecting distant metastases. It is also valuable in assessing biliary and pancreatic abnormalities. Endoscopic ultrasound

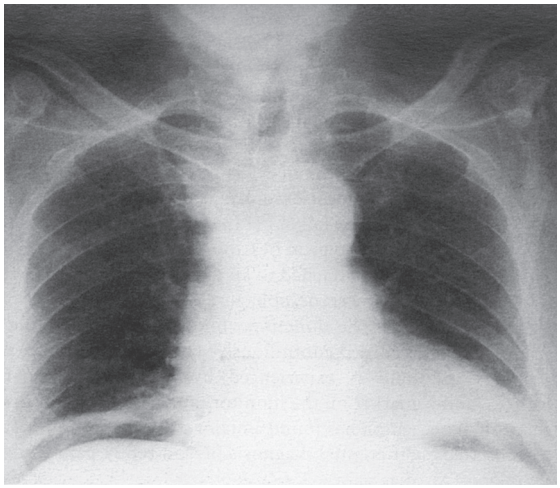


Figure 14.39 Plain X-ray of the chest showing gas under the right and left diaphragms after perforation of the duodenal ulcer. The patient was admitted in shock with abdominal pain and abdominal rigidity.

has the advantage of guiding fine needle aspiration cytology as well as other endoscopic interventions.

Percutaneous transhepatic cholangiography

Percutaneous transhepatic cholangiography complements the use of ERCP in patients with jaundice caused by obstruction of the main bile ducts, although it is usually only possible if the intrahepatic bile ducts are seen to be dilated on ultrasound. The site of the obstruction owing to tumours on the head of the pancreas or benign and malignant bile duct strictures can be localized and differentiated accurately. This technique is usually only used if ERCP fails. It also has a therapeutic function. Transhepatic drains can be placed to treat cholangitis and sepsis; stents can be placed to relieve obstruction; gallstones can be removed; and wires can be passed into the duodenum to facilitate stenting at ERCP.

Imaging (see Box 14.10)

Plain radiographs

Plain radiographs of the chest and abdomen with the patient supine and in the erect position are of great value in cases of suspected peritonitis owing to a perforated viscus, when gas may be seen under the diaphragm, usually on the right side (Fig. 14.39). A plain abdominal X-ray is useful in suspected intestinal obstruction or ileus when dilated bowel loops may be seen with fluid levels appearing on the erect X-ray (Fig. 14.40). It is important to use a plain abdominal X-ray to exclude toxic megacolon in fulminant colitis.

Abdominal ultrasound

In ultrasound scanning a probe emitting ultrasonic pulses is passed across the abdomen. Echoes

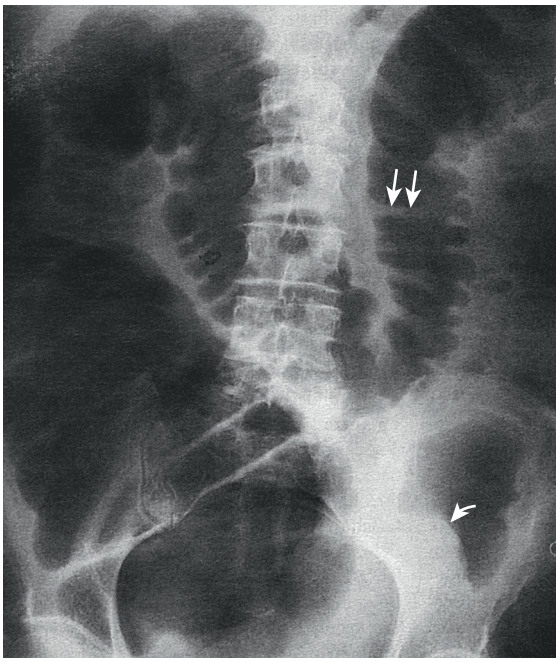


Figure 14.40 Plain X-ray of the abdomen. Obstruction of the large bowel owing to carcinoma of the sigmoid colon. Most of the colon is dilated with gas, indicating obstruction, but there appears to be no gas below the sigmoid region.

detected from within the patient are received with a transducer amplified and suitably displayed. Ultrasound examination of the liver, including Doppler assessment of the portal vein, remains the most useful and first-line imaging for all patients suspected to have a liver problem. It provides an assessment for the size and appearance of the liver, including the presence of focal liver lesions and for assessing the presence of liver cirrhosis. The ultrasound features that suggest liver cirrhosis include coarse echotexture with irregular liver outline, presence of splenomegaly, ascites and presence of reverse flow (flow away from liver-retrograde) on Doppler of portal vein. It is also useful for assessment of jaundice to assess the cause of jaundice as hepatic or obstructive from a pancreatic or biliary pathology. Patients with liver cirrhosis are at an increased risk of developing primary liver cancers known as hepatocellular carcinoma and require six-month surveillance using liver ultrasound. The main drawbacks with ultrasound are limitations of image quality owing to body shape, especially in those with obesity. It is also helpful in screening of focal liver lesions and is helpful in the diagnosis of fluid-filled lesions, such as cysts and abscesses. Liver biopsies and fine-needle aspiration of suspicious lesions can be undertaken under direct ultrasound guidance for cytology and for drainage of fluid, bile or pus. The gallbladder is most easily investigated by ultrasound. It appears as an echo-free structure. If stones are present, they are usually easily seen as mobile and echo-dense with a characteristic

'acoustic shadow' behind them. Ultrasound detects 95% of gallbladder stones, but only about 50% of stones in the bile ducts themselves. Ultrasound is particularly valuable in detecting dilation of the bile duct that may be owing to partial or complete obstruction by gallstones or tumour. Ultrasound has replaced oral cholecystography for imaging the gallbladder. Ultrasound is the usual initial technique for investigating the pancreas, is particularly useful in the diagnosis of true and pseudopancreatic cysts and is an essential tool for percutaneous needle biopsy. However, images may be obscured by overlying bowel gas and the tail of the pancreas is often poorly visualized. Ultrasound is used extensively for examining other intra-abdominal, pelvic and retroperitoneal organs and increasingly for imaging the bowel in inflammatory conditions such as in Crohn's disease. Blood flow patterns (such as in the portal vein) can be assessed by ultrasound using the Doppler principle.

Transient liver elastography

Transient liver elastography, an important tool for non-invasive assessment of liver fibrosis and quantification of liver steatosis (fat content), has reduced the need for liver biopsy. It measures liver stiffness by measuring speed of shear waves generated by ultrasound beams, with a higher score suggestive of an increase in liver stiffness and fibrosis (Fig. 14.41). Confounding factors, including obesity, concurrent significant liver inflammation, co-existent biliary dilatation and medical conditions such as heart failure and amyloidosis, may influence the accuracy. This tool is being used to screen patients who may have significant liver fibrosis/liver cirrhosis and require long-term follow-up with a liver specialist.

Computed tomography

Computed tomography is used to produce cross-sectional images of the liver and other intra-abdominal and retroperitoneal organs. It is particularly helpful in the accurate staging of all types of GI cancers, including cancers of the oesophagus, stomach, pancreas and colon (Fig. 14.42). Because it can be combined with injection of vascular contrast, it can be helpful in assessing intra-abdominal vascular abnormalities. It can facilitate guided biopsy of abnormalities and drainage of fluid and other collections. Also, increasingly it is being used to image the bowel, particularly if there is suspected obstruction, and is widely used in assessing patients with an acute abdomen. Clinicians should be aware of the radiation exposure associated with CT scanning and caution should be exercised for younger patients in particular. After scanning a patient with some gas put in the colon, the CT computer can be used to reconstruct a very accurate virtual image of the colon (CT virtual colonoscopy).

Positron emission tomography (PET) scanning

Radiolabelled isotopes that are taken up by cell metabolism are injected intravenously and the

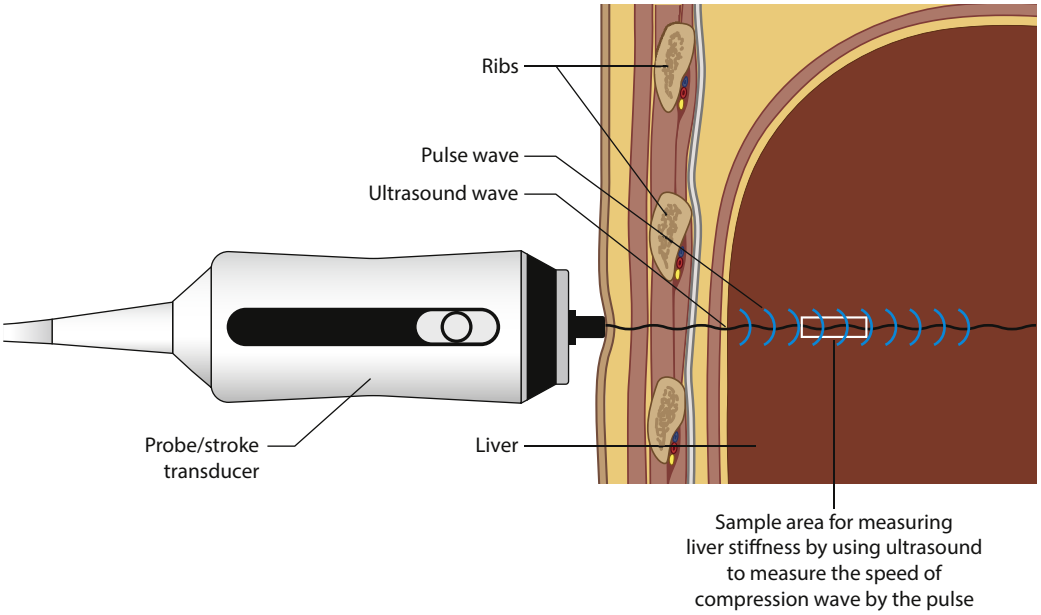


Figure 14.41 Transient liver elastrography. The probe delivers a pulse wave which spreads through the liver parenchyma. The speed of propagation of this pulse wave within the liver parenchyma is measured by the ultrasound, and in the absence of confounding factors is reflective of the degree of liver fibrosis. The measurements are expressed as kPa with <5 kPa indicating an absence of fibrosis, and >13 kPa suggesting established cirrhosis. The range from 5–13 kPa represents different fibrosis stages and can vary according to a number of variables including the underlying cause of liver disease, model of machine and other coexisting conditions such as heart failure, body habitus, liver inflammation, anti-viral therapy.

radiation that is detected is reconstructed into a cross-sectional image by the same techniques as for CT. It is a particularly sensitive technique for staging cancers, if baseline CT images are acquired at the same time

Magnetic resonance imaging (MRI and MRCP)

Magnetic resonance scanning is now replacing barium radiology for assessing the small and large bowel and perianal disease, especially when evaluating areas of stricturing and inflammation in patients with Crohn's disease and ulcerative colitis. Magnetic resonance cholangiopancreatography (MRCP) is used as a second-line imaging tool for patients with suspected biliary and pancreatic disease because it can provide high-quality images of the bile duct and pancreatic duct. The safety of this non-invasive technique has reduced the need for more invasive and hazardous examinations, such as endoscopic retrograde cholangiopancreatography (ERCP), which is reserved for patients requiring therapeutic intervention.

Magnetic resonance imaging, especially using contrast such as gadolinium, is a key tool to characterize focal liver lesions, including haemangiomas, adenomas, focal nodular hyperplasia and both primary and secondary liver cancers.

Barium studies

Barium study investigations involve administering radioopaque barium solution with direct observation of its passage using fluoroscopy through different parts of the GI tract. This study used to be first line; however, with the recent advances in the endoscopic tools, these investigations are being used less often as they involve radiation exposure and do not provide the opportunity of tissue diagnosis. Below is a brief summary of the different types.

Barium swallow

The direct observation of the passage of a radiopaque barium solution through the pharynx and oesophagus into the stomach remains an important investigation of dysphagia, especially in patients where no endoscopic diagnosis is evident, and especially where disorders of motility are suspected. Lack of progress of barium through an apparently normal lower oesophageal sphincter on swallowing in a patient with dysphagia usually indicates the presence of achalasia, a disorder of neuromuscular coordination of the oesophageal body and failure of the lower sphincter to relax.

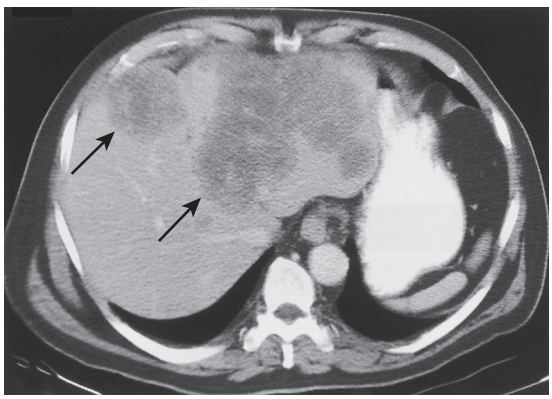


Figure 14.42 CT demonstrating liver metastases from colonic carcinoma. Large, lobulated, non-homogeneous masses (*arrows*) replace most of the left lobe of the liver.

Barium follow-through study

The small intestine may be studied by taking sequential X-rays of the abdomen after swallowing barium contrast. Abnormalities in the transit time to the colon and in small bowel pattern (e.g. dilation, narrowing, increase in transverse barring or flocculation) may be demonstrated in malabsorptive states. Areas of narrowing with proximal dilatation, fistulae and mucosal abnormalities may be produced by Crohn's disease. Small bowel diverticula or neoplasms may also be demonstrated. The amount and type of barium used for examination of the stomach/duodenum and of the small bowel is different, and so a barium follow-through should be requested as a specific examination and not as an add-on to a barium meal. However, this has been replaced largely by MRI assessment of the small bowel in view of the radiation exposure.

Small bowel enema

Small bowel enema is an alternative to the barium follow-through examination; it involves intubating the duodenum and passing small quantities of a non-flocculating barium suspension down the tube. This method is useful for detecting isolated focal lesions and strictures, but in most centres has been replaced by MR enterography and/or enteroscopy.

Barium enema

This contrast study was used to assess for colonic pathology and requires a full bowel preparation prior to barium enema (Figs 14.43 and 14.44).

Patients often find the preparation and procedure uncomfortable and there is a very small risk of perforating the colon. The inability to take diagnostic samples and the advent of CT virtual colonoscopy means that barium enemas are completely obsolete anywhere where CT scanning is available.

Other diagnostic and therapeutic investigations (see Box 14.10)

This section summarizes a battery of tests when a particular site of disease is suspected.

Investigation for liver diseases

Box 14.13 summarizes the key investigations for patients with liver disease. Most of these have

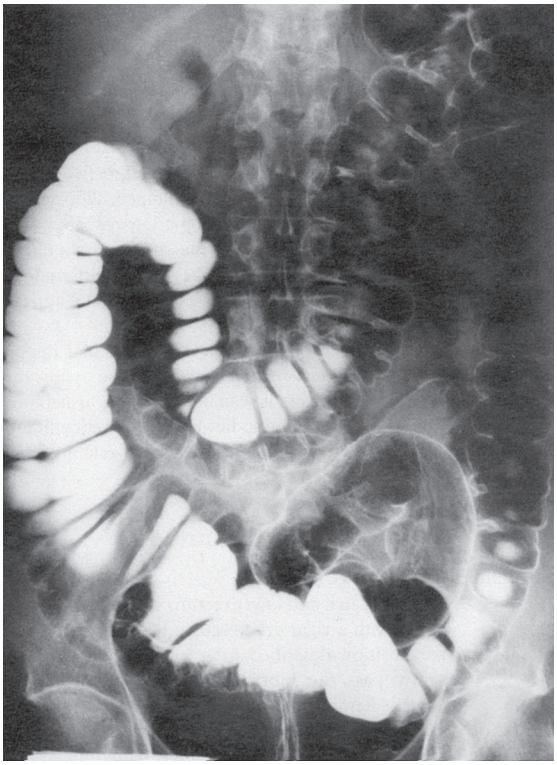


Figure 14.43 Barium enema with air contrast. The right colon is outlined by barium sulphate and the rectum, left colon and part of the transverse colon are outlined by air, with a thin mucosal layer of barium sulphate. Note the normal haustral pattern in the colon and the smooth appearance of the rectum. The anal canal can also be seen.

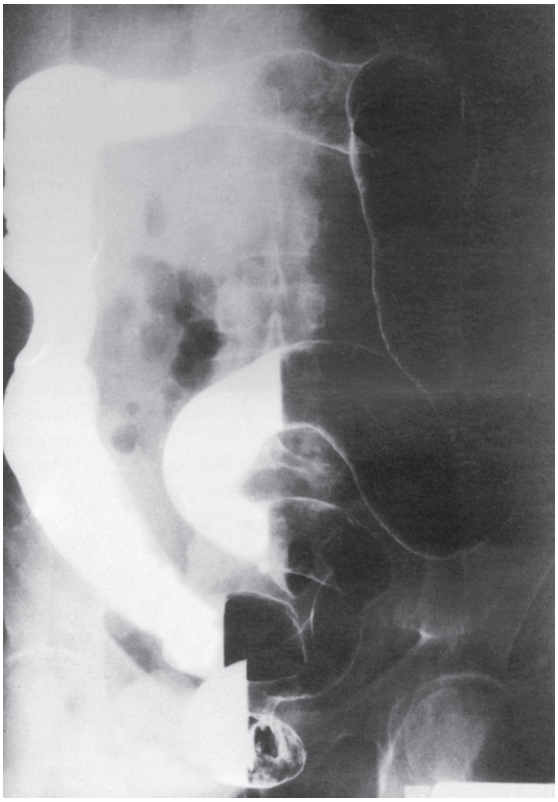


Figure 14.44 Barium enema with air contrast. In this patient with ulcerative colitis, the normal mucosal pattern and the haustra themselves have been obliterated. The patient is lying on his right side so that there are clear fluid levels in the barium sulphate suspension in the bowel.

been covered in preceding sections. However, the following additional investigations may be required in some patients.

Liver biopsy

Liver biopsy is the gold standard for confirming the nature and assessing the severity of liver disease. Percutaneous needle biopsy is the standard technique for obtaining liver tissue for histological examination, under ultrasound guidance usually under local anaesthesia (sometimes with mild sedation). Although the method is safe and reliable, there is a small but definite mortality from the procedure owing to leakage of bile and/or blood into the peritoneal cavity from the puncture site. Therefore, the procedure should always be regarded as a potentially dangerous investigation for which there should be a clear beneficial indication for the patient and should be performed by an appropriately trained operator. Contraindications to percutaneous liver biopsy include patients with abnormal clotting times or thrombocytopaenia, obstructive jaundice or ascites. Patients with abnormal clotting and ascites can have liver biopsy via transjugular route under fluoroscopic guidance by a trained operator.

Box 14.13 Investigations for assessment of liver diseases

1. Liver function tests (LFTs)
2. Liver screen
3. Liver ultrasound and Doppler of portal vein
4. Transient liver elastography
5. MRI liver (gadolinium contrast)
6. Contrast-enhanced CT liver
7. Liver biopsy
8. Hepatic venography, transjugular biopsy and transjugular intrahepatic porto-systemic shunt

Hepatic venography, transjugular biopsy and transjugular intrahepatic porto-systemic shunt

This is a radiological intervention used for undertaking liver biopsies where the percutaneous route may be high risk owing to the presence of deranged coagulation or presence of ascites. It can also help confirm portal hypertension by enabling measurement of wedged hepatic vein pressures. It has a therapeutic role as it allows placement of transhepatic portosystemic shunts (TIPS; Fig. 14.45). This is used to treat patients with variceal bleed owing to liver cirrhosis and portal hypertension that have failed endoscopic therapy or control of ascites not responsive to routine management with

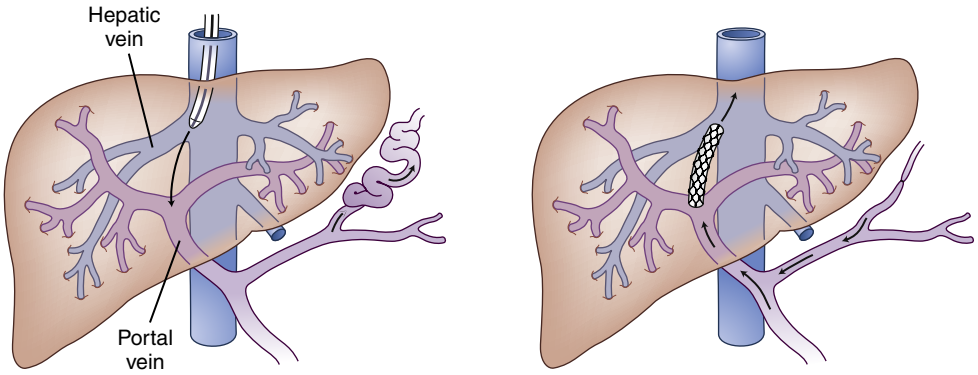


Figure 14.45 TIPS: Transjugular intrahepatic portosystemic shunt. (From Townsend C, Beauchamp RD, Evers BM, Mattox K, eds. Sabiston Textbook of Surgery: The Biological Basis of Modern Surgical Practice, ed 21. Fig. 54.21. 2022. Elsevier Inc.)

diuretics. It involves placing a stent within the hepatic parenchyma that links a branch of portal vein with hepatic vein thereby off-loading the elevated portal pressures into the systemic circulation. This helps to reduce the portal hypertension and its sequelae, including variceal bleed and ascites.

Investigation for pancreatic disease

The most specific blood test for acute pancreatitis is serum lipase, although it is not very helpful in chronic disease. The mainstay to diagnose pancreatic diseases is imaging, including abdominal ultrasound, contrast CT scan and MRCP.

Pancreatic function tests are now rarely used. In suspected pancreatic malabsorption, a therapeutic trial of pancreatic supplementation may be the easiest confirmatory test. If a definitive test is required, the pancreolauryl test can be used, a test which relies on the urine collection of a substance that can be absorbed only after GI breakdown by pancreatic enzymes. In chronic pancreatitis, faecal elastase may be reduced.

Investigation for oesophageal motility disorders

Oesophageal function studies are specialized tests indicated for patients suspected to have disorders of oesophageal motility. These include manometric studies and pH monitoring. Manometric studies of pressure changes during swallowing are used to localize functional abnormalities in the coordination of oesophageal peristalsis, for example, achalasia and oesophageal spasm. In patients with epigastric and retrosternal discomfort (heartburn) related to eating or to lying supine, reflux of acid from the stomach into the lower oesophagus should be suspected. Acid reflux can be detected best by monitoring the pH in the lower oesophagus during a 24-hour period. The pH measuring probe, passed via the nose, is placed in the lower oesophagus, 5 cm above the oesophagogastric junction, for 24 hours and the pH recorded continuously. The patient indicates when pain is experienced by pressing an electronic marker on the recorder to see if there is a correlation with the degree of acidity. This investigation has found a major clinical application in the differential diagnosis of acid reflux pain from atypical cardiac pain. A combined pH and impedance recording can also detect non-acid reflux.

Investigation for *Helicobacter pylori*

The role of infection by *H. pylori* in the pathogenesis of gastric and duodenal ulceration is well documented. *H. pylori* is a gram-negative spiral bacillus. The organism is found in the gastric antrum in about 60% of patients with gastritis or gastric ulceration and in almost all patients with duodenal ulceration. There are many asymptomatic carriers in the general population. Patients carrying *H. pylori* in the stomach will have IgG antibodies in the serum, but as these will remain present for a long period after successful treatment, serological testing is no longer recommended.

- Stool tests for *H. pylori* antigen, now the most widely used method in the community and before endoscopy, are indicated or performed.
- Endoscopic diagnosis of *H. pylori* involves carrying out a rapid urease test on a sample of gastric biopsy, referred to also as a CLO (campylobacter-like organism) test. As *H. pylori* is rich in urease, a gastric biopsy is placed in contact with a pellet or solution containing urea and a coloured pH indicator. The colour of the substrate changes when the pH is greater than 6, indicating the conversion of urea to ammonia by urease in *H. pylori*. Gastric biopsies may also be sent for culture and sensitivities testing and are indicated in those patients who do not respond to simple *H. pylori* eradication therapies.
- 13C-labelled urea breath testing, a variant of this method, utilizes 13C-labelled urea given to the patient by mouth. The patient's breath is monitored for labelled carbon dioxide, indicating breakdown of the ingested urea by urease-containing organisms in the upper GI tract. 13C-labelled urea breath testing should be used to confirm successful eradication. *H. pylori* can be rendered undetectable, but not eradicated by acid-suppression therapy, which should be stopped 2 weeks before any testing for reliable results (other than histology of gastric biopsies).

Investigation for complicated GI ulcer disease

The mainstay of investigation in such patients in endoscopy was covered in the preceding section. The following additional investigations may be required, in a sub-group of patients, depending on clinical presentations:

- Serum gastrin measurement is carried out in patients with complicated gastric and duodenal ulcer disease where a diagnosis of Zollinger-Ellison syndrome is suspected. Measurement of serum gastrin is made fasting, after a long period without acid-suppression therapy and collected into special preservative. In patients with Zollinger-Ellison syndrome the level of gastrin in the serum is increased above 100 ng/l. Serum gastrin levels are also increased in renal failure, pernicious anaemia, after vagotomy and during acid-suppression therapy.

Investigation for ongoing GI bleed

- Selective angiography of gastrointestinal arteries: This is carried out using CT-guided angiography by trained interventional radiologists in the investigations of patients presenting with ongoing haematemesis or melaena when gastroscopy and colonoscopy fail to identify the bleeding source. It is most useful if performed when the patient is actively bleeding, but may be of value if an aneurysm or abnormal tumour vasculature is present, and it can occasionally detect angiomas. It is an invasive technique that demands a great deal of skill on the part of the radiologist. It is possible to use therapeutic embolization to treat the side of bleed identified on angiography.
- Isotope scanning: Technetium-labelled red blood cells can be used to detect the location of sources of bleeding in the GI tract.

Investigation to localize inflammation

- Radioisotope studies: Technetium-labelled white cells can be used to localize inflamed bowel segments in inflammatory bowel disease; both the small and large bowel can be imaged with the same test. Isotopes taken up in the bile (hepatobiliary iminodiacetic acid, HIDA) can give images of the gallbladder and biliary tree and some information as to the functioning of these organs, such as in patients with typical biliary pain but a normal gallbladder on ultrasound examination.